Autonomous Articulated Vehicle Control via Deep Reinforcement Learning and High-Dimensional Perception

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Author's Declaration

I hereby declare that I am the sole author of this thesis. This is a true copy of the thesis, including any required final revisions, as accepted by my examiners.

I understand that my thesis may be made electronically available to the public.

Abstract

[TODO: Abstract. One or two paragraphs: the problem (learned control of an articulated tractor-trailer, including the unstable reverse manoeuvre), the approach (TD3 with matched observation and reward ablations, and an image-to-geometry perception route), the main results, and the sim-to-real deployment on the Agilex Hunter SE. To be drafted.]

Acknowledgements

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Chapter 1

Introduction

1.1 Motivation

The increasing demand for autonomous logistics and the growth of large-scale goods transportation have positioned articulated tractor-trailer vehicles as a critical target for autonomous control research. The transition from rigid-body autonomous navigation to the control of articulated tractor-trailer systems represents a significant escalation in mechanical complexity, perception requirements, and control instability, particularly during the low-speed reverse manoeuvring that dominates confined distribution-yard and loading-dock operations, which are the practical setting this thesis targets.

A tractor-trailer vehicle system is distinguished from a single-body vehicle by the articulation degree of freedom at the hitch, whose dynamics fundamentally alter the stability characteristics of the system. In reverse motion, the articulation angle γ between the tractor and trailer exhibits open-loop instability. Without precise closed-loop control, any perturbation grows unbounded, culminating in the well-known jackknife condition from which no steering correction can recover the trailer's alignment. Classical approaches such as Nonlinear Model Predictive Control (NMPC) can manage this instability by optimizing over a prediction horizon [7], but their computational demands and reliance on accurate system models limit their applicability in unstructured or sensor-rich environments.

This motivates learning-based controllers that map sensor observations directly to actuation at fixed computational cost. Their practical value, however, ultimately rests on a question that a purely simulated study cannot answer on its own, whether a policy trained entirely in simulation can be transferred to, and run on, a physical articulated vehicle. This thesis pursues both halves of that question, the design and evaluation of the learned controller in simulation, and its deployment on real hardware.

1.2 Problem Definition

The core challenge addressed in this thesis is twofold. First, the kinematic and dynamic complexity of tractor-trailer systems makes it difficult to design reward functions and control policies that simultaneously optimize path tracking performance and hitch stability. Second, real-world autonomous deployment requires perception of lane boundaries and obstacles directly from high-dimensional sensor data, rather than from pre-computed symbolic state representations.

Existing learning-based path-following studies on rigid-body vehicles have shown that continuous-control reinforcement learning can achieve performance competitive with classical baselines under nominal conditions, but robustness degrades when the vehicle model, operating conditions, or path geometry become more demanding. Those limitations are amplified for articulated systems, where reverse instability and hitch-angle growth impose a much narrower margin for control error. [TODO: need references for this paragraph]

1.3 Thesis Contributions

This thesis builds on two pieces of the author's prior graduate coursework, [TODO: how do I gracefully talk about myself?] which established its modelling and learning foundations. The first is a Deep Deterministic Policy Gradient [TODO: Cite?](DDPG) path-following study for a single-body vehicle in a custom OpenAI Gymnasium [TODO: Cite?] simulator, carried out jointly with R. Tymkow [27]. The second is a single-author linear-quadratic-regulator design for the linearized tractor-trailer model [5]. The present work combines and substantially extends these along three axes: from a single rigid body to an articulated tractor-trailer, from a hand-engineered state vector to high-dimensional perception, and adding the physical hardware deployment that neither prior report attempted.

Building on that foundation, this thesis addresses these challenges through the following contributions:

- developing a kinematic and dynamic model of a tractor-trailer system, including hitch instability analysis for reverse manoeuvring;
- designing a vision-based RL observation space that encodes a bird's-eye-view (BEV) image, rendered from a custom Pygame simulation environment, into the vehicle's path geometry through a frozen privileged-distillation encoder;
- engineering reward functions that balance cross-track error, heading error, hitch articulation stability, and collision avoidance;

- implementing and benchmarking Twin Delayed Deep Deterministic Policy Gradient (TD3) [26] agents in lane-following and obstacle-avoidance tasks within a custom Gymnasium-based simulation environment;
- comparing the learned policies against tuned classical control baselines, including pure-pursuit [18], PID, and MPC [6], on matched articulated path-tracking tasks;
- deploying the trained forward and reverse TD3 policies directly on a physical articulated platform (an AgileX Hunter SE) through a stripped-down Autoware stack (the "RL bridge") supported by an onboard LiDAR-based hitch-angle estimator that requires no trailer-mounted sensing and a Smith-predictor compensator for actuator delay (chapter 6).

The research aligns with the objectives of the University of Waterloo Mechatronics Vehicle Systems Lab (MVSL) and Electrans Technologies Ltd.

1.4 Institutional Context

This work is submitted in partial fulfilment of the requirements for a Master of Applied Science (MASc) degree in the Department of Mechanical and Mechatronics Engineering at the University of Waterloo. The degree requires six terms of full-time registration, completion of 3–5 graduate-level courses, appointment of two additional faculty readers, and a mandatory 15-day public display period. The target completion date is early July, with a seminar scheduled for late July.

Table 1.1: MASc thesis requirements summary

Element	Requirement
Research Scope	Substantial problem extending beyond existing laboratory results
Literature Review	Comprehensive assessment of classical control and modern AI approaches
Validation	Simulation-based empirical evidence across multiple environment configurations, complemented by a physical hardware deployment
Length/Format	Typically 80–120 pages; GSPA formatting and UWSpace submission
Committee	Supervisor plus two faculty readers; formal display period

1.5 Thesis Organization

The remainder of this thesis is structured as follows. Chapter 2 reviews classical and learning-based control methods for articulated vehicles, alongside the relevant perception, sim-to-real, and safety literature. Chapter 3 derives the tractor-trailer model, describes the custom Pygame/Gymnasium simulation environment, and formulates the control problem as an MDP. Chapter 4 presents the TD3 learning framework, observation spaces, reward design, representation-learning variants, and training methodology. Chapter 5 presents and discusses the simulation experiments. Chapter 6 carries the learned controller onto a physical tractor-trailer platform, describing the stripped-down Autoware/ROS2 integration that runs the policies, the onboard LiDAR hitch-angle estimator, and the Smith-predictor delay compensation. Chapter 7 summarizes the findings and outlines future work.

Chapter 2

Literature Review

Controlling an autonomous tractor-trailer system can be done through two distinct methods, classical model or optimization based or learning-based planning and control. This chapter reviews both, with the addition of sim-to-real transfer and safety considerations that bound any physical deployment. This review begins with the control problem that makes articulated vehicles distinct, surveys the classical methods that have traditionally addressed it and the learning-based methods that now challenge them, examines how perception, the simulation-to-reality gap, and the governing safety requirements shape a learned policy, and closes by identifying the specific gap that this thesis fills.

2.1 The Articulated-Vehicle Control Problem

Autonomous driving is conventionally decomposed into a pipeline of perception, planning, and control, with each stage feeding the next [1]. Within that pipeline the tractor-trailer is a demanding case for the control stage, because of the passive hitch that couples its two bodies.

Prior analyses have established why. The hitch adds an internal configuration variable, the articulation angle, whose reverse dynamics are open-loop unstable. In forward motion a small hitch disturbance decays as the tractor pulls the trailer back into alignment, but in reverse the sign of the longitudinal velocity flips this behaviour. A small articulation error grows rather than decays and, without corrective steering, runs away to the jackknife condition, a folded configuration from which no steering input can recover the trailer [2]. Reverse driving is additionally non-minimum-phase, in that to move the trailer to one side the tractor must first steer to the other [3]. These two properties, open-loop instability and non-minimum-phase response, are what separate articulated control from ordinary rigid-body lane keeping. The kinematic and dynamic model that produces them is derived in chapter 3, and the confined low-speed setting that fixes the capabilities a controller must deliver is set out in chapter 1.

2.2 Classical Model-Based Control

The longest-established response to articulated instability is model-based control, deriving a dynamic model, then designing a control law that provably stabilizes it. This section reviews the two layers of that approach, the feedback controllers that stabilize and track, and the motion planners that supply the reference for them to track. Together they form the conventional pipeline against which the learned controller of this thesis is benchmarked.

2.2.1 Feedback Control and Stabilization

At the simplest level, a proportional-integral-derivative (PID) controller is trivial to implement and computationally negligible, but it acts only on the present error and cannot anticipate the jackknife condition. Its gains must also be re-tuned for different payloads and path distributions. A linear state-feedback law derived from the linearized tractor-trailer model addresses the multi-variable coupling more directly. Choosing that feedback as a linear-quadratic regulator (LQR) trades tracking accuracy against control effort through a quadratic cost and is optimal for the linearized plant [4]. The author's prior work applied exactly this design to the linearized tractor model [5].

Model predictive control (MPC) generalizes linear feedback by re-optimizing the steering sequence over a receding horizon subject to explicit state and input constraints [6]. This constraint handling is valuable for articulated vehicles because a hard bound can be placed directly on the articulation angle. Linear MPC suffices near an operating point, whereas nonlinear MPC (NMPC) retains the full nonlinear kinematics, including the reverse instability, at a higher computational cost [7]. Control tailored specifically to the articulated case has taken two further forms. Active trailer steering adds an actuated degree of freedom at the trailer axle or hitch to damp articulation directly, and has been shown to improve both the high-speed stability and the low-speed manoeuvrability of tractor-semitrailers [8,9]. For a conventional trailer with no such actuation, all authority rests on the tractor's steering angle, and recent work has used an intrinsically stable MPC formulation to suppress jackknifing while tracking a reference, exploiting stable inversion of the non-minimum-phase dynamics [10].

2.2.2 Geometric and Nonlinear Control

A parallel line of work exploits the geometric structure of the non-holonomic system directly. Trailer systems are a classic example of a differentially flat system, meaning that the full state and the control inputs can be written algebraically in terms of a flat output and its derivatives, which reduces trajectory generation to the problem of shaping that output [11]. Converting the kinematics to chained form enables time-varying feedback laws with provable path-following and point-stabilization guarantees for wheeled vehicles [12]. For the specific and difficult case of reverse motion, feedback schemes have been derived that stabilize a truck and trailer backing along a path in spite of the open-loop instability [3]. These methods achieve strong guarantees by committing fully to an analytical model of the vehicle, which is precisely the commitment that the learned controller of this thesis seeks to avoid.

2.2.3 Motion Planning for Articulated Vehicles

The feedback controllers above track a reference path, which must first be planned. For articulated vehicles this planning problem is itself difficult, because a feasible path must respect the vehicle's non-holonomic constraints and avoid configurations from which the trailer cannot recover. Sampling-based planners such as the rapidly-exploring random tree and its asymptotically optimal variant RRT* explore the configuration space without an explicit discretization [13,14], whereas search-based freespace planners such as hybrid A* discretize the search yet retain kinematic feasibility [15]. Tailored to the articulated case, lattice-based planners assemble precomputed motion primitives that connect circular-equilibrium configurations, enabling complex forward and reverse manoeuvres [16], and optimization-based formulations compute reference paths that minimize the swept area of the vehicle bodies subject to road and obstacle constraints [17]. Once a path exists, geometric trackers such as pure pursuit follow it by steering toward a look-ahead point on the path [18], and refined trackers such as the Stanley controller add an explicit cross-track error term with demonstrated performance on full-scale autonomous vehicles [19].

2.2.4 Strengths and Limitations

The appeal of this pipeline is that it comes with guarantees. Given an accurate model, the stability of the controller and the feasibility of the planned path can be analyzed and, in principle, guaranteed. Its cost is precisely that dependence. The model must be identified and kept accurate across payloads and operating conditions, the controllers and planners must be tuned, and the system must be assembled from separately designed planning and control stages. These costs motivate the alternative pursued in this thesis, a controller learned directly from interaction, which maps observations to steering commands without an explicit model inversion or a separately planned reference trajectory, though the path to be followed still enters through the policy's observation, and which is benchmarked in chapter 5 against tuned instances of exactly the classical controllers reviewed here.

2.3 Learning-Based Control

Where a classical controller encodes the vehicle model explicitly and executes a fixed control law, a reinforcement learning (RL) agent infers a control policy directly from interaction with its environment, optimizing the policy end-to-end against a scalar reward [20]. This trades the analytical guarantees of the model-based approach for the ability to learn behaviours that are difficult to hand-design and to fold perception and control into a single learned mapping. This section traces the RL methods relevant to the thesis from their general foundations to their specific application to articulated vehicles.

2.3.1 Deep Reinforcement Learning Foundations

Modern RL for control rests on the deep-network function approximators that made high-dimensional value estimation practical. The Deep Q-Network first showed that a single architecture could learn to act directly from high-dimensional visual input across a range of tasks, using experience replay and a target network to stabilize learning [21]. Value-based methods of this kind are naturally suited to discrete action sets. The continuous steering and speed commands of a vehicle instead call for policy-based methods.

2.3.2 Actor-Critic and Policy-Gradient Methods

The deterministic policy gradient provides a route to continuous control by learning a deterministic actor alongside a critic that estimates action value [22]. Deep Deterministic Policy Gradient (DDPG) is the deep-network realization of this idea and a canonical off-policy actor-critic algorithm for continuous action spaces [23], but it is prone to critic overestimation bias and unstable updates. Two lines of work address this. Proximal Policy Optimization (PPO) is an on-policy method that improves stability through a clipped objective, at the cost of lower sample efficiency, which matters when simulation rollouts are expensive [24]. Soft Actor-Critic (SAC) instead augments the reward with an entropy term to encourage exploration and improve robustness in continuous control [25]. This thesis adopts Twin Delayed DDPG (TD3), a direct successor to DDPG that targets its two weaknesses through twin critics, delayed actor updates, and target-policy smoothing [26], and whose role is detailed in chapter 4. The author's prior coursework used DDPG for single-body path following [27]. The move to TD3 here is a direct response to the training instability observed in that work.

2.3.3 Reinforcement Learning for Autonomous Driving

RL has been applied across the driving stack, and recent surveys catalogue its use for lane keeping, behavioural decision making, and end-to-end control [28]. Two threads are relevant here. The first is end-to-end learning, in which a single network maps sensor input to control, whether trained by imitation of a human driver [29] or by reinforcement. A notable demonstration of this trained a lane-following policy on a full-scale car with on-board RL in a single day [30]. The second is the explicit framing of the driving task as a Markov decision process with a shaped reward, as set out in early RL-for-driving frameworks [31]. Both threads inform the observation and reward design developed in this thesis.

2.3.4 Reinforcement Learning for Articulated Vehicles

Applying RL specifically to articulated vehicles is more recent but growing, and the earliest instance long predates deep RL. The neural truck-backer-upper of Nguyen and Widrow learned to reverse a trailer toward a target, and depended on training that exposed the network to the unstable configurations it had to correct [32]. Modern work has revisited the problem with deep methods. A DDPG backing controller paired with a fuzzy-logic supervisor learned to reverse while avoiding the jackknife state [33], and policy-gradient agents have been trained to track reference paths with a tractor-trailer using rewards that penalize both tracking deviation and articulation growth [34]. A recurring theme in this small body of work, and one this thesis examines directly through its reward and observation ablations, is the importance of the hitch-stability term. The reward must discourage articulation growth for reverse driving to be learned reliably. Reward shaping of this kind rests on a firm theoretical footing, as potential-based shaping can accelerate learning while provably preserving the optimal policy [35].

2.3.5 Sequence Models and Curriculum Learning

Beyond actor-critic control, Decision Transformers recast RL as sequence modelling, predicting an action from a context window of past states, actions, and returns-to-go [36]. Such models can represent longer temporal dependencies than a shallow actor-critic policy, but their performance depends heavily on the diversity of the offline dataset, and a dataset dominated by nominal tracking behaviour tends to give poor recovery from the high-error states it underrepresents, an instance of the distribution-shift problem that motivates deliberately collecting such states. This need for coverage of hard configurations reappears as a training-data problem regardless of the policy class, and it motivates curriculum learning: presenting tasks in a deliberately ordered sequence, or automatically emphasizing the situations the agent handles worst [37, 38]. This thesis retains an actor-critic formulation but adopts a failure-replay curriculum (chapter 4) that revisits failed scenarios, an idea related in spirit to prioritized experience replay [39].

2.3.6 Learning under Actuation Delay

A policy trained against an idealized, instantaneous actuator can degrade when deployed on hardware whose steering and drive systems exhibit dead time and first-order lag. A constant actuation delay violates the Markov property of the underlying decision process, turning it into a delayed MDP whose optimal policy depends on the recent action history rather than the instantaneous state alone [40]. The learning literature offers two broad remedies: augmenting the state with a buffer of in-flight actions so that the delayed problem is again Markovian, and correcting the off-policy value estimates for the delay, as in the Delay-Correcting Actor-Critic [41]. This thesis instead takes a complementary, model-based route at deployment, namely a Smith-predictor compensator [42] that presents the delay-free-trained policy with a delay-compensated state estimate (chapter 6), which avoids retraining the policy against the physical delay.

2.4 Perception and Representation Learning

A learned controller is only as good as what it perceives. A central question of this thesis is whether a compact, hand-engineered state vector or a learned representation of a bird's-eye-view (BEV) image yields better and more transferable control. This section reviews the perception building blocks that make the latter possible.

2.4.1 Convolutional Networks for Driving

Convolutional neural networks (CNNs) are the standard front-end for visual perception, learning spatial features directly from images rather than from hand-specified descriptors [43, 44]. Network depth was later shown to be trainable at scale through residual connections, which underpin most modern vision backbones [45]. In driving, CNNs have been trained end-to-end to map camera images directly to steering commands [46]. The bird's-eye view is a particularly convenient representation for ground-vehicle control because it places the vehicle and its surroundings in a top-down frame. Recent perception systems learn to lift camera images into exactly such a representation [47], and the classical occupancy grid is its long-standing hand-built analogue [48]. In this thesis a CNN processes a rendered BEV image to extract lane geometry and obstacle layout.

2.4.2 Representation Learning and Autoencoders

Training a control policy directly on raw images is sample-inefficient, because the policy must learn to see and to act at the same time. Autoencoders decouple these by compressing images into a compact latent code in an unsupervised pre-training stage [49], an idea applied to RL to shrink the policy's input and accelerate learning [50]. Variational autoencoders impose a probabilistic structure on the latent space that can improve its regularity [51], and spatial autoencoders tailored to control learn latent features corresponding to task-relevant image locations [52]. The U-Net architecture, originally developed for image segmentation, preserves multi-scale spatial detail through skip connections and provides a stronger structural prior than a plain convolutional encoder [53]. This thesis treats these learned encoders as one end of a representation spectrum, with the engineered state vector at the other, and compares them directly in chapter 4.

2.4.3 Range Sensing and Point-Cloud Perception

Bird's-eye imagery is not the only high-dimensional modality available to a ground vehicle. Planar lidar provides a direct geometric measurement of the surrounding free space, which the classical occupancy grid represents as a discretized map of obstacle probability [48]. At the other extreme of complexity, deep networks such as PointNet learn features directly from raw, unordered three-dimensional point clouds [54]. This thesis deliberately adopts a lighter-weight lidar representation that sits between these poles, a fixed-length vector of range returns appended to the state vector, rather than a learned point-cloud encoder. This keeps the observation compact and the sensing model simple, and it allows the lidar and image modalities to be compared on an equal footing in chapter 4.

2.5 Sim-to-Real Transfer

Because the controllers in this thesis are trained entirely in simulation, transferring a simulation-trained policy to physical hardware is a first-class concern rather than an afterthought. The central obstacle is the reality gap, the systematic mismatch between simulation and hardware that degrades a naively transferred policy [55]. It is useful to separate this gap into a visual component, a difference in what the sensors report, and a dynamics component, a difference in how the vehicle responds to a command, because the two are attacked by different tools.

The dominant learning-based remedy is domain randomization, which randomizes parameters during training so that the real system appears to the policy as one more variation of a distribution it has already learned to handle. It has been applied to the visual gap by randomizing rendering [56] and to the dynamics gap by randomizing physical parameters such as mass and friction [57]. Randomization can also be closed into a loop. System-identification methods adapt the simulator's parameter distribution from a small amount of real experience, so that simulated and real behaviour are matched rather than merely broadened [58], while domain-adaptation methods instead learn a mapping that reconciles simulated and real observations [59]. A complementary strategy is to raise simulator fidelity directly. The open urban driving simulator CARLA is a widely used example [60], and chapter 6 describes a geographically faithful simulator built for this work. Actuation delay is a particularly damaging instance of the dynamics gap for articulated reversing, and is addressed both by the delay-aware learning methods discussed above and by the Smith-predictor compensator of chapter 6. That chapter characterizes the specific gaps encountered on the physical platform, namely actuation delay, hitch-angle sensing noise, and path-distribution mismatch, and addresses them through targeted engineering compensation while noting domain randomization as a route to further robustness.

2.6 Safety and Standards

A controller intended for a physical vehicle must ultimately answer to safety requirements, and the learning-based setting raises safety questions of its own. Safe reinforcement learning studies how to learn while respecting constraints or avoiding catastrophic states [61], and control barrier functions offer a complementary, model-based mechanism to overlay hard safety constraints on a learned policy, a direction identified here as future work [62]. For an articulated vehicle the binding safety concern is the jackknife condition, which the reward must discourage and the operational limits of chapter 6 enforce. A fielded deployment would in addition have to sit within the established automotive-safety framework, most directly the Safety of the Intended Functionality standard for hazards that arise under nominal operation [63], alongside the broader functional-safety, safety-case, and automation-level standards [64–66]. These requirements bound the present simulation study rather than being evaluated within it, and they frame the deployment reported in chapter 6.

2.7 Synthesis and Research Gap

The literature reviewed in this chapter divides into a mature model-based tradition and a fast-moving learning-based one, and the gap this thesis addresses lies between them. Classical control offers guarantees but demands an accurate model, careful tuning, and a separate planning stage. Learning-based control removes those demands but must contend with reward design, the reverse instability, high-dimensional perception, and the reality gap. To our knowledge, prior learned controllers for articulated vehicles have been demonstrated almost exclusively in simulation and on a single, hand-specified observation modality. The backing controllers of Nguyen and Widrow [32] and Bejar and Morán [33], and the tracking agent of Kang et al. [34], each learn from a fixed state representation and report results in simulation, without a physical deployment or a comparison across perception modalities. This thesis addresses that combination directly. It develops a learned articulated-vehicle controller with high-dimensional perception, compares observation modalities and reward formulations through controlled ablations, benchmarks the result against tuned classical baselines, and deploys the trained policies on physical hardware with explicit attention to hitch stability and sim-to-real transfer. The remainder of the thesis carries out this programme, beginning with the vehicle model and simulation environment in chapter 3.

Chapter 3

Modelling, Simulation, and Problem Formulation

3.1 Tractor Dynamics

The tractor is represented by a planar single-track bicycle model [67], in which the front axle is lumped into one steerable wheel and the rear axle into one fixed wheel. Its global position (x, y) follows from the body-frame longitudinal and lateral velocities (v_x, v_y) and the heading ψ:

$$\dot{x} = v_x \cos\psi - v_y \sin\psi \tag{3.1}$$

$$\dot{y} = v_x \sin\psi + v_y \cos\psi \tag{3.2}$$

Lateral force is generated at each tire as a function of its slip angle. Under the small-angle assumptions appropriate to the operating envelope, the front and rear slip angles are

$$\alpha_f = \delta - \frac{v_y + l_f \dot{\psi}}{v_x}, \qquad \alpha_r = -\frac{v_y - l_r \dot{\psi}}{v_x}, \tag{3.3}$$

where δ is the steering angle. Rather than evaluating the full Pacejka magic-formula tire curve, the simulator operates in its linear region. For the modest slip angles encountered here the curve is well approximated by a constant cornering stiffness, giving the lateral tire forces

$$F_{yf} = C_f\,\alpha_f, \qquad F_{yr} = C_r\,\alpha_r. \tag{3.4}$$

This linearized-Pacejka (constant cornering-stiffness) model is the one implemented in the simulator. It preserves the lateral behaviour relevant to path tracking while avoiding the hard-to-identify parameters of the full nonlinear tire curve. Collecting the longitudinal, lateral, and yaw-moment balances, with a quadratic aerodynamic drag $F_{\text{drag}} = \frac{1}{2} C_d \rho A v_x^2$ and a net drive force F_x, yields the tractor dynamics. Although the drag force is negligible at low speeds, it was added as a damping force so that the simulation eventually came to rest under no input. Active braking is achieved through applying a negative drive force.

$$m\left(\dot{v}_x - v_y\dot{\psi}\right) = F_x - F_{\text{drag}} - F_{yf}\sin\delta, \tag{3.5}$$

$$m\left(\dot{v}_y + v_x\dot{\psi}\right) = F_{yf}\cos\delta + F_{yr}, \tag{3.6}$$

$$I_z\,\ddot{\psi} = l_f F_{yf}\cos\delta - l_r F_{yr}. \tag{3.7}$$

For control design and for the time-stepping used in simulation, these equations are linearized about straight-line motion ($\delta \approx 0$) at a constant reference speed $v_x = v$. With state $\boldsymbol{\xi} = [\,v_x,\ v_y,\ \dot{\psi}\,]^\top$ and input $\mathbf{u} = [\,F_x,\ \delta\,]^\top$, the linearized dynamics take the time-invariant form $\dot{\boldsymbol{\xi}} = A\boldsymbol{\xi} + B\mathbf{u}$ with

$$A = \begin{bmatrix} -\dfrac{C_d\rho A v}{2m} & 0 & 0 \\ 0 & -\dfrac{C_f + C_r}{mv} & \dfrac{l_r C_r - l_f C_f}{mv} \\ 0 & \dfrac{l_r C_r - l_f C_f}{I_z v} & -\dfrac{l_f^2 C_f + l_r^2 C_r}{I_z v} \end{bmatrix}, \qquad B = \begin{bmatrix} \dfrac{1}{m} & 0 \\ 0 & \dfrac{C_f}{m} \\ 0 & \dfrac{l_f C_f}{I_z} \end{bmatrix}. \tag{3.8}$$

The continuous system is converted to the discrete update $\boldsymbol{\xi}_{k+1} = A_d\boldsymbol{\xi}_k + B_d\mathbf{u}_k$ by a zero-order-hold discretization at the simulation step $\Delta t = 0.1\,\text{s}$, and it is this discrete map that is advanced at each timestep.

This level of fidelity is sufficient for the control tasks studied in this thesis, where the quantities of interest are path-relative position error, heading error, steering angle, and hitch articulation. The aim is a physically meaningful environment for training and evaluating classical and trained policies, not a high-fidelity vehicle simulator. An overly simplified kinematic model of the tractor would make the simulation task unrealistically easy and could reduce transferability to physical systems by neglecting actuator response, steering-rate limits, and tire-force effects on the vehicle's motion. In contrast, a kinematic model was chosen for the trailer because, at the low speeds considered in this thesis, the trailer's dominant behaviour is governed by its rolling constraint rather than by lateral tire-slip or inertial dynamics. The full-length trailer therefore primarily contributes geometric articulation through the hitch. Its axle constrains the trailer to roll along its longitudinal direction, and this nonholonomic constraint determines the evolution of the hitch angle during forward and reverse manoeuvres. This reduced-order representation preserves the main source of reverse instability while avoiding the additional tire, load-transfer, and yaw-inertia parameters required by a full dynamic trailer model.

3.2 Trailer Articulation and Reverse Instability

The articulated system adds a trailer body coupled to the tractor through a revolute hitch. The relevant geometry is:

- L_h: signed distance from the tractor rear axle to the hitch point (in both the training model and the hardware platform the hitch sits at the rear axle, so $L_h \approx 0$);
- L_t: distance from the hitch to the trailer axle, i.e. the trailer wheelbase;
- $\gamma = \psi - \psi_t$: the articulation angle between the tractor heading ψ and the trailer heading ψ_t (the same convention used by the deployment in chapter 6).

The hitch point is carried rigidly by the tractor,

$$x_h = x - L_h \cos\psi, \qquad y_h = y - L_h \sin\psi, \tag{3.9}$$

and translates with the tractor's longitudinal speed v. Imposing a pure-rolling (no lateral slip) constraint on the trailer axle fixes the trailer yaw rate from the lateral component of the hitch velocity:

$$\dot{\psi}_t = \frac{v}{L_t}\sin(\psi - \psi_t) = \frac{v}{L_t}\sin\gamma. \tag{3.10}$$

The articulation angle therefore evolves as

$$\dot{\gamma} = \dot{\psi} - \frac{v}{L_t}\sin\gamma, \tag{3.11}$$

where the tractor yaw rate $\dot{\psi}$ is supplied by the tractor model of eq. (3.8) (equivalently $\dot{\psi} = (v/L)\tan\delta$ for a purely kinematic tractor of wheelbase $L = l_f + l_r$). The trailer axle pose then follows from the hitch point and the trailer heading,

$$x_t = x_h - L_t \cos\psi_t, \qquad y_t = y_h - L_t \sin\psi_t. \tag{3.12}$$

The sign of v in eq. (3.11) is what makes reverse motion qualitatively harder. Linearizing about the aligned configuration ($\gamma \approx 0$, so $\sin\gamma \approx \gamma$) and holding the steering-driven tractor yaw fixed, the unforced articulation mode obeys

$$\dot{\gamma} \approx -\frac{v}{L_t}\gamma, \qquad \lambda = -\frac{v}{L_t}. \tag{3.13}$$

In forward motion ($v > 0$) the eigenvalue λ is negative. Articulation errors decay and the trailer self-aligns behind the tractor. In reverse ($v < 0$) the eigenvalue becomes positive and the same errors grow exponentially toward the jackknife condition. This is an open-loop instability rather than a mere tuning difficulty, and it is the defining feature of reverse articulated manoeuvring, the reason the reverse task demands active hitch stabilization, trailer-referenced error metrics, and the dedicated reward and curriculum treatments developed in chapter 4.

The jackknife toward which this instability drives is a folded configuration from which no steering input can recover the trailer's alignment, so successful reverse manoeuvring demands anticipatory control that corrects a developing articulation error before it grows large enough to become unrecoverable. In practice the controller must regulate three coupled objectives at once (trailer path tracking, tractor placement, and articulation stability), and a controller that attends only to the tractor's cross-track error can look reasonable in the short term while driving the hitch toward instability.

3.3 Vehicle Parameters

The vehicle model is parameterized to represent a shunt truck (terminal tractor, or yard hostler), the class of vehicle that performs the confined low-speed trailer manoeuvring in distribution centres that this thesis targets. The specific reference vehicle is the Kalmar Ottawa T2 4x2 terminal tractor. Its layout maps directly onto the single-track bicycle model of eq. (3.8), and its manufacturer specification sheet provides genuine primary-source data [68]. The tractor has a wheelbase of 2.95 m, a maximum steering angle of 50° (0.87 rad), and is governed to a top speed near 20 m/s, more than enough to operate in the low-speed regime studied here. Table 3.1 lists the parameter set used in simulation, corresponding to the bobtail (no-trailer) diesel configuration.

Table 3.1: Vehicle parameters used in simulation, for the bobtail Kalmar Ottawa T2 diesel terminal tractor.

Parameter	Value	Parameter	Value
m	6577 kg	l_f	1.33 m
I_z	16 000 kg m^2	l_r	1.62 m
C_f	190 000 N/rad	C_d	0.8
C_r	200 000 N/rad	A	7.0 m^2
Δt	0.1 s	ρ	1.225 kg/m^3

The mass, wheelbase, and overall dimensions are taken directly from the manufacturer specification sheet. The yaw moment of inertia I_z is estimated from the mass and body envelope using a standard moment-of-inertia estimator for road vehicles [69]. The cornering stiffnesses C_f and C_r are derived from published heavy-truck tire data [70], scaled by the number of tires per axle (a single pair at the steered front axle against dual tires at the drive axle) and by the axle loads. The front and rear stiffnesses therefore differ, unlike the equal-stiffness simplification commonly used for passenger-car examples. The longitudinal centre-of-gravity split is estimated from an approximate static front/rear axle load distribution, and the aerodynamic terms C_d and A are engineering estimates for a terminal-tractor cab. Neither the aerodynamic terms nor the centre-of-gravity split is backed by primary data, but their influence is minor in the operating regime of interest. Aerodynamic forces are negligible at governed yard speeds, and the centre-of-gravity split mainly shapes high-speed transient response rather than the low-speed trajectories studied here. The timestep $\Delta t = 0.1\,\mathrm{s}$ corresponds to the 10 Hz control loop used in the hardware deployment (chapter 6), so the simulation and the deployed controller operate at a matched rate.

The values in table 3.1 describe the bobtail tractor, which defines the dynamic tractor subsystem used in simulation. The trailer is represented kinematically through the hitch model of eq. (3.11), for the reasons given in section 3.1. Its inertia is deliberately omitted rather than approximated. Because the tasks prescribe longitudinal speed, trailer mass never enters the control loop through the longitudinal balance, and both the rolling constraint of eq. (3.10) and the reverse-instability eigenvalue of eq. (3.13) are independent of mass and moment of inertia. Trailer inertia is therefore not a control-critical state in the nominal environment. The one regime in which a heavily loaded trailer does reshape the dynamics, through kingpin load transfer onto the tractor, is examined separately as a robustness variant, in which that load is folded into the tractor's mass, yaw inertia, cornering stiffness, and centre-of-gravity split rather than introduced as a separate trailer body.

3.4 Simulation Environment

The training and evaluation environment is a custom 2D simulation built on the Farama Foundation Gymnasium library [71]. Gymnasium provides a standardized API for environments to integrate directly with reinforcement learning algorithm libraries, including Stable Baselines3 [72].

The simulation is implemented in Python, with Pygame [73] providing collision masks and rendering. It was built around a rendering pipeline that draws the tractor-trailer and lane geometry as a top-down frame. This representation can be viewed live for monitoring, or passed directly to the image encoder as a bird's-eye-view (BEV) observation, making high-dimensional perception available to the agent without a separate image-synthesis step. At each timestep, the discrete-time tractor-trailer state-space equations are re-evaluated from the previous state and control inputs, updating the vehicle position, heading, hitch angle, and associated error signals, and Pygame redraws the scene as a 2D top-down view from which the BEV image observation is cropped. For large-scale training this rendered observation is later produced without the Pygame renderer by the batched GPU simulator (section 4.9), but the rendering pipeline remains the conceptual origin of the BEV observation and the reference for what it encodes.

The simulation environment includes:

1. **Vehicle Model**: A combined dynamic tractor (bicycle model with a linearized Pacejka tire model [74]) and kinematic trailer with hitch constraints, using the shunt-truck parameters of section 3.3.
2. **Lane Generation**: Randomly generated lane paths using cubic spline interpolation, producing varying curvature paths for training and evaluation.
3. **Obstacle Placement and Local Re-planning**: Randomly placed static obstacles within the lane and a local path planner that perturbs the centreline around those obstacles for avoidance experiments.
4. **Collision Detection**: Pygame sprite mask-based collision detection, providing pixel-precise contact sensing between the vehicle body and obstacles or lane boundaries.
5. **Observations**: the rendered top-down BEV image described above, alongside engineered state and range-sensor (lidar) observations. The full set of observation modalities available to the policy is defined in chapter 4.

3.5 Environment Tasks

The environment implementation is structured around reusable task and observation components. A base lane-following task is combined with different observation heads, while obstacle-aware variants reuse the same task logic through an environment that adds obstacle generation, local path planning, and obstacle-aware reward evaluation. This avoids maintaining separate forward, reverse, lidar, and BEV implementations with duplicated task logic.

3.5.1 Line Following

The lane-following environment extends the base tractor-trailer environment to present a lane-tracking task. A reference lane is generated at the start of each episode using cubic spline interpolation through randomly placed waypoints. The reward function encourages the agent to keep both the tractor and trailer within the lane boundaries while maintaining progress along the path.

Reverse line-following environments use the same path generation and rendering pipeline, but reverse the longitudinal command and compute reward and tracking terms with the trailer as the leading unit, so that the same task can be studied in backward manoeuvres without changing the task definition.

3.5.2 Obstacle Avoidance

The obstacle avoidance environment adds static obstacles within the lane. The agent must navigate the tractor-trailer through the lane while avoiding obstacles, which requires both path-following behaviour and collision avoidance. Instead of following the global lane centreline directly, the environment constructs a local path around the placed obstacles and evaluates tracking error relative to this locally planned path. Obstacles are placed as a single cluster of one or two circular obstacles per episode, drawn from a small set of layout categories (clear, shift, squeeze, and blocked), so that the difficulty distribution is controlled and can be stratified during evaluation. The locally planned avoidance path is used only to shape the reward and is never exposed to the agent, so the detour must be inferred from the lidar or BEV observation. In addition to terminal collision penalties, the base environment applies a per-step proximity penalty near blocked occupancy-grid cells, so both lane boundaries and inserted obstacles discourage the agent before contact occurs. Forward and reverse obstacle environments are both available. The scalar difficulty rating that indexes these layouts, and the reward that keys on it, are developed in section 4.3.1.

3.5.3 Actions and Episode Termination

The policy is exposed to a deliberately narrow interface. In the reinforcement-learning sense its action is two-dimensional, a continuous steering rate $\dot{\delta}$ and a stop signal σ:

$$\mathbf{a} = [\dot{\delta},\, \sigma] \in [-25°/\mathrm{s},\, 25°/\mathrm{s}] \times [-1,\, 1]. \tag{3.14}$$

Longitudinal speed is not itself a control degree of freedom. The environment is driven by a target speed, but the policy does not set it directly. Instead, a thin interface layer maps the stop signal onto that speed, commanding the fixed operating value of the current task (5 m/s forward, −5 m/s in reverse) while the vehicle drives and zero speed the instant a stop is issued. Because speed is prescribed in this way, every method solves the same lateral problem and the forward and reverse tasks share a common control interface. Commanding the steering rate rather than the steering angle reflects the fact that the steering angle cannot change instantaneously on a physical vehicle, so regulating its rate of change is more physically realistic. The stop signal is a go/stop decision. When σ exceeds a fixed threshold the vehicle halts and the episode terminates, which lets the policy decline to attempt a layout it judges impassable. This stop-capable action is used across all learning-based tasks, so that the interface does not change between the no-obstacle and obstacle settings. Discrete algorithms realize the same choice through a joint steering-and-stop action set. Stopping is not universally desirable. In the obstacle-free tasks a stop is treated as a failure and penalized, and it earns reward only on difficult obstacle layouts. The difficulty-keyed stop-gate reward that governs this trade-off is developed in section 4.3.1, and the experimental protocols in chapter 5.

Episodes terminate on collision, jackknife ($|\gamma| > 90°$), or successful path completion.

Chapter 4

Learning Framework: Observation and Representation Design, Reward Formulation, and Policy Training

4.1 Learning Framework Overview

This chapter discusses the development the learned controller and the supporting framework required to train it. The core learning system is defined first: a Twin Delayed Deep Deterministic Policy Gradient (TD3) agent that operates on a compact engineered state and is driven by a reward that balances path tracking against hitch stability. The higher-dimensional observation modalities, planar lidar and the bird's-eye-view image, and the obstacle-aware navigation task are then introduced as extensions of that core system. Because inserting raw perception into an otherwise successful policy turns out to destabilise learning, a dedicated section then develops the training methodology that makes high-dimensional-observation policies trainable at all, working through three progressively cleaner ways to decouple perception from control and arriving at a geometry-distillation encoder that recovers from-scratch learning entirely. The evaluation methodology and the GPU-parallel simulation infrastructure that make the empirical study feasible close the chapter. This progression mirrors how the design was actually arrived at: the algorithm and reward are settled on the engineered state where learning is fast and unambiguous, then the added complexity of raw perception and the training machinery it demands is introduced.

Figure unavailable

Architecture of the learning system: an observation (engineered state, lidar, or BEV image) is encoded into a feature vector and passed to the TD3 actor and critics, which output a steering rate and a stop signal. The reward combines path-tracking and hitch-stability terms.

`figures/architecture/learning_framework`

Figure 4.1: Architecture of the learning system: an observation (engineered state, lidar, or BEV image) is encoded into a feature vector and passed to the TD3 actor and critics, which output a steering rate and a stop signal. The reward combines path-tracking and hitch-stability terms.

The agent perceives its surroundings through one of three observation families, forming a deliberate progression from a fully engineered representation to raw imagery:

- **State**: an 8-dimensional vector $[\delta, \gamma, e_y, e_\psi, e_{y,t}, e_{\psi,t}, \kappa_1, \kappa_2]$ containing the steering angle, the hitch articulation angle, the tractor and trailer cross-track and heading errors, and two local path-curvature samples.

- **Minimal State + lidar**: a 2-dimensional vector $[\delta, \gamma]$ containing the steering angle and the hitch angle augmented with N normalized range readings, where the beam count $N \in \{8, 24, 64\}$ is varied in the lidar ablation of chapter 5.

- **Minimal State + BEV image**: The same minimal state vector and a 32×32 grayscale bird's-eye-view image.

The core learning system of the next three sections operates on the engineered state. The lidar and image modalities are taken up in section 4.5. This progression lets the marginal value of richer perception be measured against its added representation-learning and inference cost rather than assumed.

The lidar and BEV based observation spaces are augmented with the minimal state vector because steering angle and hitch angle are available through onboard sensor readings and do not need to be learned by the perception system.

4.2 Reinforcement Learning with TD3

Twin Delayed Deep Deterministic Policy Gradient (TD3) is an off-policy actor-critic algorithm designed for continuous action spaces. TD3 extends deterministic policy-gradient methods with three key improvements: twin critic networks (to reduce overestimation bias), delayed policy updates (to improve stability), and target policy smoothing (to reduce variance in critic updates). These properties make TD3 a natural fit for articulated-vehicle control: the action space is continuous (a continuous steering rate and a stop signal that is triggered above a threshold), a deterministic policy avoids the action-distribution spread that can aggravate the non-minimum-phase reverse dynamics, and the off-policy replay buffer amortizes the comparatively high cost of image-based rollouts by reusing every transition many times.

The TD3 actor network $\mu(s \mid \theta^{\mu})$ maps states to deterministic actions, while the critic networks $Q_1, Q_2(s, a \mid \theta^{Q_1}, \theta^{Q_2})$ evaluate action quality. The target value is computed using the minimum of the two critic estimates:

$$y = r_t + \gamma_d \min_{i=1,2} Q_i(s_{t+1},\, \mu(s_{t+1}) + \epsilon) \tag{4.1}$$

where $\epsilon \sim \mathrm{clip}(\mathcal{N}(0, \sigma), -c, c)$ is clipped Gaussian noise added to the target action for smoothing. Parameters are updated by minimizing the temporal difference error in the critics and maximizing the expected return through the actor using the deterministic policy gradient theorem [22].

The TD3 implementation used for this work is from the Stable Baselines3 library [72], which provides a reliable implementation integrated with the Gymnasium environment API [71].

4.3 Reward Formulation

The thesis evaluates several reward variants rather than a single scalar shaping rule. Let e_y and e_ψ denote the tractor cross-track and heading errors, $e_{y,t}$ and $e_{\psi,t}$ the trailer cross-track and heading errors, γ the hitch articulation angle, $\dot{x}$ the longitudinal speed, and δ the steering angle. Longitudinal speed is not a control degree of freedom: it is held at the fixed operating value from chapter 3 ($\dot{x} = \pm 5\,\mathrm{m/s}$ forward or reverse), so $\dot{x}$ appears in the reward only as a near-constant progress term that a stop forfeits, and the second action dimension is a stop signal rather than a commanded speed (chapter 3). The pure-pursuit guide steering is denoted δ_{pp}, with steering deviation $\Delta_\delta = (\delta - \delta_{\mathrm{pp}})/(2\delta_{\max})$. The lane-following base environment also defines a proximity penalty P_{prox} that is subtracted from non-terminal step rewards. This penalty is zero when the vehicle is farther than $d_{\mathrm{prox}} = 2\,\mathrm{m}$ from blocked occupancy-grid cells and increases quadratically to $P_{\max} = 5$ as clearance approaches zero:

$$P_{\mathrm{prox}} = \begin{cases} 0, & c \geq d_{\mathrm{prox}}, \\ P_{\max}\left(1 - \frac{c}{d_{\mathrm{prox}}}\right)^2, & c < d_{\mathrm{prox}}, \end{cases} \tag{4.2}$$

where c is the minimum clearance between the rendered vehicle footprint and the lane-boundary/obstacle occupancy grid. In the no-obstacle lane-following environment this discourages driving near lane boundaries; obstacle environments inherit the same term for both lane boundaries and inserted obstacle cells.

Forward lane-following rewards. For forward driving, the terminal reward is

$$R^{\text{fwd}}_{\text{term}} = \begin{cases} 100, & \text{success}, \\ -100, & \text{failure}, \end{cases} \tag{4.3}$$

and the common shaping terms are

$$r_p^{\text{fwd}} = 0.5\,\dot{x}, \tag{4.4}$$

$$P_{\text{tractor}} = e_y^2 + e_\psi^2, \tag{4.5}$$

$$P_{\text{trailer}} = (0.5\,e_{y,t})^2 + (0.5\,e_{\psi,t})^2. \tag{4.6}$$

The four forward reward variants are then

$$R^{\text{fwd}}_{\text{dense}} = r_p^{\text{fwd}} - P_{\text{tractor}} - P_{\text{trailer}} - P_{\text{prox}}, \tag{4.7}$$

$$R^{\text{fwd}}_{\text{tractor}} = r_p^{\text{fwd}} - P_{\text{tractor}} - P_{\text{prox}}, \tag{4.8}$$

$$\begin{aligned} R^{\text{fwd}}_{\text{mult}} &= 4\exp(-|e_y| - 0.5|e_\psi| - 0.75|e_{y,t}| - 0.5|e_{\psi,t}|)\exp(-1.5|\gamma|) \\ &\quad + 0.5\,\mathrm{clip}(\dot{x}, 0, 1) - P_{\text{prox}}, \end{aligned} \tag{4.9}$$

$$R^{\text{fwd}}_{\text{guide}}(k) = \alpha_k\left(-5\Delta_\delta^2 - P_{\text{prox}}\right) + (1-\alpha_k)\,R^{\text{fwd}}_{\text{mult}}. \tag{4.10}$$

The guided reward interpolates between a pure-pursuit imitation term $-5\Delta_\delta^2$, which rewards matching the pure-pursuit steering, and the multiplicative reward $R^{\text{fwd}}_{\text{mult}}$ defined above. The schedule $\alpha_k \in [0, 1]$ governs the transition from imitation toward the task reward and is described in section 4.6, where the same mechanism is reused to fine-tune an imitated policy. At $\alpha_k = 1$ the guided reward is a pure pure-pursuit clone.

Reverse lane-following rewards. For reverse driving, the terminal reward is

$$R^{\text{rev}}_{\text{term}} = \begin{cases} 200, & \text{success}, \\ -500, & \text{failure}, \end{cases} \tag{4.11}$$

with shared shaping terms

$$r_p^{\text{rev}} = 3\,\mathrm{clip}(-\dot{x}, 0, 1), \tag{4.12}$$

$$P_{\text{path}} = e_y^2 + 0.5\,e_\psi^2 + 0.5\,e_{y,t}^2 + 0.25\,e_{\psi,t}^2, \tag{4.13}$$

$$P_{\text{jack}} = 10\max(0, |\gamma| - 0.4)^2. \tag{4.14}$$

The reverse variants are

$$\begin{aligned}
R^{\text{rev}}_{\text{dense}} &= r^{\text{rev}}_p - P_{\text{path}} - P_{\text{jack}} - P_{\text{prox}}, && (4.15)\\
R^{\text{rev}}_{\text{no-hitch}} &= r^{\text{rev}}_p - P_{\text{path}} - P_{\text{prox}}, && (4.16)\\
R^{\text{rev}}_{\text{mult}} &= 5\exp(-|e_y| - 0.5|e_\psi| - 0.75|e_{y,t}| - 0.5|e_{\psi,t}|)\exp(-2|\gamma|)\\
&\quad + 0.5\,\text{clip}(-\dot{x}, 0, 1) - P_{\text{prox}}, && (4.17)\\
R^{\text{rev}}_{\text{guide}}(k) &= \alpha_k\left(-5\Delta_\delta^2 - P_{\text{prox}}\right) + (1-\alpha_k)\,R^{\text{rev}}_{\text{mult}}. && (4.18)
\end{aligned}$$

Obstacle difficulty. The obstacle-aware terms below are keyed to a scalar difficulty $d \in [0,1]$, a static geometric clearance ratio computed once per episode at the tightest blocked cross-section:

$$d = \text{clip}\left(1 - \frac{g_{\max} - w_v}{w_L - w_v},\ 0,\ 1\right), \tag{4.19}$$

where $g_{\max}$ is the widest contiguous free lateral gap, w_v the vehicle width, and w_L the lane width, so $d = 0$ is a fully open lane and $d = 1$ a gap at or below the vehicle width. Its rationale and per-direction calibration are developed in section 4.3.1 and chapter 5.

Obstacle-aware extensions. Obstacle environments inherit the underlying forward or reverse reward, including the proximity penalty above. A failed obstacle episode receives the standard terminal penalty (chapter 3). A deliberate stop instead receives a difficulty-gated terminal reward,

$$\begin{aligned}
s(d) &= \frac{e^{k_d d} - 1}{e^{k_d} - 1}, \qquad k_d = 3, && (4.20)\\
R^{\text{obs}}_{\text{stop}} &= s(d)\,B_{\text{hard}} - \big(1 - s(d)\big)B_{\text{easy}} + p\,B_{\text{prog}}, && (4.21)
\end{aligned}$$

where $p \in [0,1]$ is the progress made before stopping and $B_{\text{hard}} = B_{\text{easy}} = 60$, $B_{\text{prog}} = 20$. The soft difficulty gate $s(d)$ makes stopping pay on an impassable layout, where it must beat a crash, and lose to driving through on an easy one.

Branching hidden-path reward. Rather than tracking the single hidden avoidance path of chapter 3, the obstacle reward tracks two hidden corridors, one passing each side of the obstacle, and keeps the more favourable of the two evaluations:

$$R^{\text{branch}} = \max_{c\,\in\,\{\text{L},\text{R}\}} R\left(e_y^{(c)},\, e_\psi^{(c)},\, e_{y,t}^{(c)},\, e_{\psi,t}^{(c)}\right), \tag{4.22}$$

where $e_\bullet^{(c)}$ are the tracking errors relative to corridor c and $R(\cdot)$ is the underlying forward or reverse reward above. Its rationale is developed in section 4.3.1.

The variants above differ chiefly in how the tracking and articulation terms combine, additively or as a product that forces the errors down jointly rather than letting the agent collect partial credit from a single term, and are compared empirically in chapter 5.

4.3.1 Obstacle-Navigation Reward Design

The obstacle-navigation task reuses the lane-following reward but adds three design elements, defined in the equations above and motivated here: a difficulty rating, a stop-gate, and a branching hidden-path reward. The obstacle environment itself was described in chapter 3.

Difficulty as a geometric proxy. The difficulty d of eq. (4.19) is a deliberately cheap, static geometric proxy for drivability, used because evaluating true kinematic feasibility for a non-holonomic tractor-trailer inside the training loop would be far too slow. Measuring only whether the body width fits the gap has two consequences that matter later: every gap at or below the vehicle width collapses to $d = 1$, and, because the proxy ignores the manoeuvring and trailer-swing room a real trajectory needs, it labels forward and reverse layouts identically even though reverse needs far more room. Calibrating d into a per-direction drivability threshold is a result of chapter 5.

Stop-gate. The difficulty that gates the stop is windowed rather than a single global per-episode scalar. It is computed over only the obstacles that fall inside the observable window ahead of the vehicle, and it is supplied to the policy as an observation channel, so the stop decision is a causal function of what the agent can currently see rather than of a hidden per-episode label. Keying the stop reward on this same windowed difficulty makes the value of a stop a deterministic function of the observation, which is what lets the policy learn a clean stop decision. The difficulty channel sets the prize for a correct stop, while perception sets its timing and the progress term credits driving through as much of the layout as possible first. The reverse task needs a further refinement, a state-dependent version of this windowed difficulty that rises as the trailer tracks off the avoidance path, which is developed with its results in chapter 5.

Branching hidden-path reward. Rewarding agreement with a single prescribed avoidance corridor would penalize the agent for choosing an equally valid other side. The branching reward of eq. (4.22) removes this by crediting the nearer of two hidden corridors, so the side of passage becomes an emergent decision from perception. This ties the obstacle task to the observation-space study. Choosing the feasible side and knowing when to stop both require seeing far enough ahead.

4.4 Hyperparameter Optimization

The TD3 agent requires careful hyperparameter tuning. The actor-critic learning rates, replay-buffer size, target-network update rate, and exploration-noise scale are selected over repeated simulation rollouts. TD3 optimization uses step-based updates so that the replay buffer and gradient schedule remain consistent across single-environment and vectorized runs. The throughput of vectorized and GPU-batched execution is addressed in section 4.9. The same backbone and hyperparameters are shared across the state, lidar, and BEV observation variants wherever possible, so that the observation ablation isolates the effect of the observation and representation rather than of the training configuration. This sharing cannot extend to the algorithm comparison of chapter 5, where TD3, SAC, PPO, and DQN each require the settings appropriate to their own update rule (for example, on-policy batch and rollout lengths for PPO, or the exploration schedule and discretization for DQN). There, only the environment, reward, and observation are held fixed, and each algorithm is given its own sensible configuration so that the comparison reflects the algorithms rather than a single tuning that happens to favour one of them.

4.5 Perception and Representation Design

The core learning system above operates on the engineered state vector. This section introduces the two higher-dimensional observation modalities that this thesis compares against that baseline, a planar lidar scan and a bird's-eye-view image, and the representation learning required to turn raw imagery into features the policy can use.

4.5.1 Lidar Observation

The obstacle environments expose a simulated planar lidar observation by appending normalized obstacle distances to the state vector. In forward driving the lidar pose is attached to the tractor, while in reverse driving it is attached to the trailer and rotated by π so that the sensing direction follows the leading unit. This design keeps the sensing model consistent with the physical task. The front of the articulated system changes when the direction of travel reverses. The lidar is a qualitatively different modality from the image, a sparse one-dimensional range scan rather than a two-dimensional picture, and the beam count is treated as a resolution knob in the ablation of chapter 5.

4.5.2 Bird's-Eye-View Observation

The bird's-eye-view (BEV) observation is a 32×32 single-channel top-down image of a local region (approximately a 20 m square) centred on the vehicle and oriented so that the direction of travel points toward the top of the image. It encodes the task-relevant spatial structure that the state vector cannot: lane geometry, the local path, and obstacle layout. Forward and reverse variants share the same size and format and differ only in the crop direction (the forward observation biases the crop toward the upcoming path, the reverse observation toward the region behind the vehicle) and in the interpretation of the accompanying state vector, which uses trailer-referenced tracking errors in reverse because the trailer leads the manoeuvre.

As introduced in section 3.4, the BEV image originates from the Pygame renderer, which draws the top-down scene as a grayscale frame that is cropped to the local window. For large-scale training the same observation is produced without the renderer by the batched GPU simulator (section 4.9), which rasterizes it analytically as an occupancy image, where each pixel records whether the corresponding ground point is drivable (inside the lane corridor and clear of obstacles) or blocked. The two share resolution and role. The analytic form omits photometric detail and does not draw the vehicle body, which sits at the image centre by construction.

Graded (signed-distance) encoding. A binary occupancy image is a poor learning target even though it is a faithful map. It is flat almost everywhere, so the task-relevant quantity, the clearance between the vehicle and the nearest boundary, is carried only by sparse gradients at the cell edges. A randomly initialized convolutional encoder reads those edges as high-frequency noise, which is one route by which raw vision destabilizes early learning (section 4.6). The BEV channel is therefore optionally rendered as a graded signed-distance field rather than a hard mask. The pixel value rises to $+1$ at the corridor centreline, falls smoothly to 0 at the drivable edge, and goes negative outside, so clearance becomes a dense, task-aligned signal available at every pixel. Two further coordinate channels in the style of CoordConv [75] can be appended so the convolution can locate a feature within the crop rather than only detect its presence. This graded representation is the image encoding carried forward into the perception ablation of chapter 5.

4.5.3 Image Encoding and Representation Learning

The BEV image must be turned into a feature vector before it can drive the policy. Two routes are studied: training an image encoder end-to-end with the agent, and pretraining an encoder so that a usable visual representation is available from the first control step.

A Convolutional Neural Network (CNN) [43] serves as the end-to-end image front-end, implemented as a custom feature extractor within Stable Baselines3 [72]. Matching the scale of the 32×32 observation, the image branch is deliberately small, a two-layer convolutional encoder with $1 \to 5 \to 5$ channels maps the image to an 80-dimensional feature vector, followed by a linear projection to 64 image features. A separate two-layer multilayer perceptron embeds the 8-dimensional state vector to 64 dimensions. The two branches are concatenated into a 128-dimensional feature vector for the TD3 actor and critic heads. Two choices in this extractor are made specifically for stability with a high-dimensional input: layer normalization is applied to the image features so their scale cannot swamp the state embedding, and the actor and critic feature extractors are kept separate, following the TD3 default, so that critic representation updates do not overwrite the actor's features.

Training this image branch end-to-end is not free of pitfalls. Inserting a randomly initialized CNN branch into an otherwise successful state-based policy can harm early TD3 learning. The agent converged to a short-horizon wall-collision behaviour even though the same vector state was sufficient for rapid learning. Two diagnostic encoders isolate this effect. The `state_only` BEV encoder keeps the dictionary observation and multi-input policy path but ignores the image branch, confirming that the BEV environment and multi-input wrapper can still learn from the vector state alone. The `scaled_cnn` encoder introduces the image branch gradually through a scalar schedule:

$$\phi(s) = [\lambda_I(k)\,\phi_I(I),\, \phi_x(\tilde{x})]\,, \tag{4.23}$$

where ϕ_I is the image encoder, ϕ_x is the vector-state encoder, and $\lambda_I(k)$ ramps from zero to one over training. A second schedule can reduce the non-steering vector components after the image branch has reached full weight:

$$\tilde{x} = [\delta,\, \lambda_x(k)\gamma,\, \lambda_x(k)e_y,\, \lambda_x(k)e_\psi,\, \lambda_x(k)e_{y,t},\, \lambda_x(k)e_{\psi,t},\, \lambda_x(k)\kappa_1,\, \lambda_x(k)\kappa_2]. \tag{4.24}$$

The steering angle δ is intentionally preserved because it is not reliably inferable from the 32×32 BEV image.

These diagnostics motivate decoupling perception from control, and this thesis pursues a progression of ways to do so, developed in section 4.6. The first and most direct decouples at the level of the representation. If a random image branch can derail learning that the vector state makes easy, then supplying a usable visual representation from the first training step should remove that failure mode. An autoencoder is therefore used to pretrain an image representation before policy optimization [50]. The encoder $E(\cdot)$ maps the image observation s to a feature vector z, and the decoder $D(\cdot)$ reconstructs the image:

$$z = E(s) \tag{4.25}$$

$$\hat{s} = D(z) \tag{4.26}$$

The autoencoder is pretrained on BEV images collected from a mixture of simple pure-pursuit rollouts, already-trained compatible policies, and random actions, to broaden state coverage. After reconstruction training, the encoder weights are transferred into the RL feature extractor and can be frozen or fine-tuned while the policy is trained.

A UNet-style [53] autoencoder provides a second, stronger structural prior. Its encoder uses repeated double-convolution blocks with max pooling, followed by global average pooling to produce a compact 128-dimensional image descriptor. During pretraining the full UNet decoder retains skip connections to improve reconstruction quality. For RL, only the encoder path is kept and concatenated with the vector embedding. It tests whether preserving multi-scale spatial detail during representation learning improves downstream control, particularly in scenes where obstacle shape and lane boundaries must both be interpreted from the image.

Together, the end-to-end CNN, the frozen and fine-tuned autoencoder encoders, and the UNet encoder form a spectrum of image representations, from learned-during-control to learned-before-control, that is compared against the engineered state and lidar in chapter 5. Pretraining the encoder is only the first of the decoupling strategies, however, and on its own it addresses only part of the problem. The stronger routes decouple at the level of the policy and of perception itself, and they rest on a specific view of why raw vision destabilizes learning. That view, and the training methodology built on it, are the subject of the next section.

4.5.4 Observation Space Summary

The full observation interface used across experiments is summarized in table 4.1.

Table 4.1: Observation space components

Component	Description	Role
`vector`	8-dimensional state: $[\delta, \gamma, e_y, e_\psi, e_{y,t}, e_{\psi,t}, \kappa_1, \kappa_2]$	Provides precise articulation and path-tracking state information
`lidar`	N normalized obstacle ranges, with $N \in \{8, 24, 64\}$	Encodes local free space around the leading unit without image processing
`image`	32×32 BEV crop, binary occupancy or graded signed-distance, with optional coordinate channels	Encodes lane layout, local path shape, obstacle positions, and vehicle geometry

4.6 Training High-Dimensional-Observation Policies

Introducing a raw image into the observation does more than add parameters. It changes what it takes to train the policy at all. The diagnostics of section 4.5 showed that inserting a randomly initialized image branch into an otherwise successful state policy can collapse learning that the vector state alone makes easy. This section first sets out a hypothesis for why that happens, then works through three progressively cleaner ways to decouple perception from control: pretraining the representation, warm-starting the policy by imitation, and distilling the state into a frozen perception module. Each route addresses more of the hypothesized failure than the last, and the third recovers ordinary from-scratch learning entirely. The hypothesis is stated here as motivation. The evidence that isolates the cause and separates the three routes is presented in chapter 5.

4.6.1 An Off-Policy Instability Hypothesis

The failure is hypothesized to be an optimization pathology of off-policy actor-critic learning rather than a limitation of perception. In TD3 the actor is trained to maximize the critic's value estimate, while the critic is trained by bootstrapping from its own estimates of the actor's next actions. Early in training the critic is essentially random, and it over-estimates the value of actions far from the data it has seen. This is extrapolation error [76], a manifestation of the deadly triad that arises when bootstrapping, function approximation, and off-policy updates are combined [20]. With a low-dimensional state the actor recovers quickly. With a high-dimensional image, the actor and its convolutional encoder have far more capacity to chase the critic into these over-valued, off-distribution actions before the critic corrects, so the policy can settle into a degenerate short-horizon behaviour. Two ideas from the literature target the same mechanism and inform the designs below: constraining the actor to stay near the behaviour data, as in the TD3+BC offline method [77], and decoupling the encoder from the reward-driven critic gradient, as in the stop-gradient image encoder of SAC-AE [78]. The stronger claim, that this mechanism and not the image content, the representation, or the encoder is the true cause, is tested directly by the causal ablation of chapter 5.

4.6.2 Route A: Pretraining the Representation

The first and most direct response, introduced in section 4.5, is to pretrain the image encoder by reconstruction (the autoencoder and UNet) so that a usable representation exists from the first control step. This attacks the noisy co-trained encoder head-on, but it addresses only one of the two failure sources: it does nothing about the actor being dragged off-distribution by a cold critic, and reconstruction spends encoder capacity on task-irrelevant pixel detail rather than on the geometry the controller needs. It is a partial fix, and it motivates the two stronger routes below.

4.6.3 Route B: Imitation Warm-Start

If a cold critic dragging the actor off-distribution is the problem, then starting from a policy that is already good, and keeping it there, should avoid it. This second decoupling strategy acts at the level of the policy rather than the representation. A competent low-dimensional `state` agent is trained first and then used as an expert. It is rolled out inside the image environment, its observations and actions are recorded, and the image policy (convolutional encoder and actor head together) is trained by supervised regression to reproduce the expert's actions. This behaviour-cloning warm-start [46] supplies a non-degenerate policy and encoder from the first reinforcement-learning step, so the actor never has to survive a cold critic from scratch. One caveat, revisited in chapter 5, is that the observation still contains the full state vector and the expert's actions derive from it, so the cloning loss can be satisfied largely through the state branch. The warm-start therefore yields a competent, non-collapsing policy, but it does not by itself prove that the convolutional encoder has learned strong image features. Whether the image supplies information the state lacks is a separate question, one that only becomes measurable on tasks where the state is genuinely insufficient, and is left to future work.

4.6.4 Route C: Geometry Distillation

The cleanest route removes both failure sources at once, by refusing to let reinforcement learning shape the encoder at all. A convolutional encoder is trained supervised to predict the vehicle's geometry directly from the bird's-eye-view image, using the simulator's ground-truth state as a privileged teacher label. This is the privileged-distillation, or learning-by-cheating, paradigm [79]: a teacher that sees the true state supervises a student that must infer it from the image. The trained encoder is then frozen and used as a fixed perception module, and the policy is trained from scratch with ordinary TD3 over a low-dimensional vector, exactly as in the engineered-state baseline.

The encoder takes the graded signed-distance BEV with its two coordinate channels (section 4.5) through a small three-layer convolutional tower into a 128-dimensional trunk with layer normalization, from which a linear head regresses the standardized geometry vector. The observation the policy receives is then split along the physics of the problem. The internal states, the steering angle δ and the hitch articulation angle γ, are supplied as true proprioception, because they are cheap direct measurements on the vehicle and, being internal, are not reliably recoverable from an ego-centric top-down image, as already noted for δ in section 4.5. The exteroceptive geometry, the tractor and trailer tracking errors and the path-curvature look-aheads, is taken from the frozen encoder's prediction, because that is precisely what an external top-down view encodes.

This design removes both failure sources identified above. Because the encoder is trained only by supervised regression and then frozen, the critic is never fed the nonstationary, high-variance features of a co-trained encoder, so the concern behind Route A does not arise. Because the policy is trained from scratch on the frozen features, its actor and critic co-evolve from a cold start, so there is no pretrained-actor-versus-fresh-critic mismatch of the kind Route B must manage. The learning regime is therefore the same as the engineered-state baseline that learns readily. Distillation also settles a question the imitation route could not: predicting the state from the image alone cannot be satisfied through a state branch, so success at the prediction task is direct evidence that the image is a sufficient statistic for the geometry it encodes. The per-component recoverability of the geometry and the from-scratch learning curve are reported in chapter 5.

4.6.5 Safe Adaptation

A warm-started policy raises a further question: can reinforcement learning then improve it without re-triggering the collapse? This is the problem of safe adaptation, and this thesis treats it as a design space rather than a solved method. The most conservative option is to freeze the actor and its encoder and train only the critic, which cannot destabilize a policy it never updates, but for the same reason cannot improve it either. Allowing the actor to move again is where the risk returns. Two principled anti-collapse designs are identified for this purpose: an action anchor that penalizes departures from the cloned policy, in the spirit of TD3+BC [77], and a frozen-base-plus-residual policy that constrains updates to a small correction around the warm-started actor. Both aim to keep the actor inside the on-distribution region where the critic's values are trustworthy. Their effectiveness, and the prior question of whether the lane-tracking task even leaves reward headroom for adaptation to capture, are examined in chapter 5. The reverse direction, which is both harder to clone and more collapse-prone, is left as a known gap.

A single annealing schedule underlies these transitions from imitation to improvement: $\alpha_k = \max(0,\, 1 - k/T)$, which decays from one to zero over T steps. It appeared already in the guided reward defined earlier in this chapter, where it blends the pure-pursuit imitation term into the multiplicative task reward, so that the agent first clones the classical controller and is then progressively rewarded for surpassing it. The same idea governs the fine-tuning of an imitated policy. Holding α_k near one keeps the agent close to its imitation while the critic warms up, and decaying it hands control to the task reward once the values are trustworthy. Fixing $\alpha_k = 1$ throughout instead recovers pure imitation, which is how the guided reward is used when a stable clone, rather than improvement, is the goal.

4.6.6 Curriculum and Training Protocol

Two further training choices apply across the obstacle experiments. First, obstacle policies are not trained from scratch but through a two-stage curriculum: Stage A trains a lane-follower in the stop-capable action space with no obstacles, and Stage B warm-starts from it with obstacles enabled, so the agent begins obstacle training already able to track a lane and operate the stop signal. The reverse Stage B is itself warm-started from a pre-validated reverse lane-follower rather than trained from scratch, because reverse learning from scratch is high-variance (chapter 5). This curriculum is distinct from the failure-replay curriculum of section 4.7, which reshapes scenario sampling within a single training run. Second, the off-policy runs use a learning rate of 10^{-3}, since raising it to 2×10^{-3} diverges once the stop action is present, together with 32 parallel environments, the regime found stable for TD3 on this task. On-policy algorithms, which consume far larger environment batches, are configured separately with the per-algorithm settings discussed in section 4.4 and enter only the algorithm comparison of chapter 5.

4.7 Failure Replay Curriculum

A persistent challenge in training RL agents on the reverse lane-following task is that the episode failure rate is high early in training and the lane geometry is re-randomized at every reset. In the standard random-reset regime, each failure is discarded and replaced by a fresh, randomly drawn layout. As a result, the agent rarely encounters the same hard configuration twice, reducing the pressure to master any particular geometry and potentially slowing convergence on the inherently open-loop-unstable reverse task.

To address this, a failure replay curriculum is implemented as a transparent environment wrapper. At each episode boundary the wrapper inspects the episode outcome. If the episode terminated due to a failure (jackknife, collision, or out-of-bounds departure before reaching the goal), the next reset reuses the same random seed that generated the current path and spawn position, exactly replaying the same scenario. If the episode was completed successfully or timed out, a fresh seed is drawn and a new random layout is used. A configurable consecutive-retry ceiling (default five retries) prevents the agent from being trapped indefinitely on a pathologically difficult layout.

This strategy is a form of automatic curriculum learning [37]: rather than constructing a manually graded task sequence, the curriculum emerges from the agent's own failure signal. Its emphasis on repeatedly revisiting the transitions the agent handles worst is shared, in spirit, with prioritized experience replay [39], though here the emphasis acts at the level of episode sampling rather than replay-buffer weighting. It is implemented orthogonally to the reward formulation and observation representation ablations. The wrapper modifies only the episode-level sampling policy, not the per-step reward signal or the information available to the agent, and is therefore controlled by a separate training flag. Models trained with and without failure replay are saved to distinct directories to ensure clean comparison.

The curriculum is most meaningful for scenarios where the per-episode failure rate is high and layout difficulty is non-uniform, namely reverse lane-following and reverse obstacle avoidance. For forward lane-following the failure rate is low after minimal training and the benefit is expected to be negligible.

4.8 Evaluation Methodology

The observation, reward, and training designs above are assessed by a common evaluation protocol, described once here and applied throughout chapter 5. Its purpose is to make the comparisons fair and the reliability of each result explicit.

Every policy is evaluated on a fixed pool of pre-generated scenarios rather than on freshly sampled episodes, so that different controllers face identical layouts. Outcomes are recorded categorically as completion, deliberate stop, collision, or timeout. Because the obstacle task's difficulty distribution is bimodal, results are reported stratified by the geometric difficulty of eq. (4.19) rather than pooled, so that easy and hard layouts are not averaged into a single misleading number. Each reported rate is a proportion, so it is given with a Wilson 95% confidence interval, and each configuration is run over multiple random seeds and reported as a mean with its across-seed interval, so that a result's stability, and not only its central value, is visible.

A raw collision rate is misleading for a policy that can decline to attempt a layout, because stopping deflates it: a policy that stops on everything never crashes. The honest avoidance metric is therefore the collision rate conditional on attempting, crash/(complete + crash), reported alongside the raw collision rate and the stop rate. The minimum clearance to any obstacle over an episode is also reported, as a continuous measure of how close the vehicle came to contact.

To interpret the stop-gate behaviour, the geometric difficulty is calibrated against actual drivability using a classical stack as an oracle. The traditional pipeline (a global path, an artificial-potential-field local planner, and a pure-pursuit tracker, the last of which is one of the classical baselines detailed in appendix A) is run with the stop action disabled so that it always attempts, and its per-difficulty completion rate is read as an empirical drivability $P_{\text{clear}}(d)$. This turns the static geometric proxy of eq. (4.19) into a measured, direction-specific quantity, and it fixes the difficulty at which stopping becomes rational. With the terminal rewards used in this comparison (success +100, collision −500), the expected value of attempting a layout is

$$\mathbb{E}[\text{attempt}](d) = 100\, P_{\text{clear}}(d) - 500\,\big(1 - P_{\text{clear}}(d)\big), \tag{4.27}$$

which is zero at $P_{\text{clear}} = 5/6 \approx 83.3\%$. The difficulty at which measured drivability crosses this break-even is the true, per-direction drivability threshold $d^\star$ against which the stop-gate policy should be judged, rather than the raw geometric scale. The forward and reverse values of $d^{\star}$, and what they reveal about the reverse task in particular, are reported in chapter 5. So that completion measures tracking rather than stop-timing, the stop action is disabled during the lane-following and drivability evaluations and enabled only when the stop-gate behaviour itself is under test.

4.9 Scalable Simulation: GPU-Parallel Batched Environments

The observation, reward, and curriculum studies described in this chapter, together with the controller benchmark of chapter 5, require training and evaluating a large matrix of agents: several observation modalities, multiple reward formulations, forward and reverse variants, and repeated random seeds for statistical validity. The scalar Pygame/Gymnasium environment of section 3.4 is the physically faithful reference model, but it advances a single environment at a time in Python and rebuilds an occupancy grid on every reset, which makes a study of this size prohibitively slow. To make the empirical campaign tractable, the environment was re-implemented as a batched simulator that advances N independent environments simultaneously on a graphics processing unit (GPU). The original scalar environment is retained unchanged as the parity oracle against which the batched implementation is validated, so that the physical fidelity established in chapter 3 continues to hold for every result reported in this thesis.

4.9.1 Batched Environment Design

The batched environment adds a leading batch axis of size N to every state array, so that a single vectorized operation advances all N environments at once. It is written against a thin array-backend abstraction and runs identically on NumPy for the CPU [80] or CuPy for the GPU [81], selected at run time by a single environment variable. Because CuPy exposes a NumPy-compatible interface, the same vectorized code executes on either device without modification. Achieving this required replacing five constructs in the scalar environment that are inherently sequential or defined one environment at a time:

1. the per-step zero-order-hold discretization, computed in the scalar model with `scipy.signal.cont2discrete`, is replaced by a batched matrix exponential using the scaling-and-squaring method [82], which discretizes all N state-space models in one operation;

2. the per-reset occupancy grid and k-d tree nearest-neighbour lookup are replaced by an analytic distance-to-centreline computation together with an on-device pool of pre-generated paths, which removes the reset bottleneck entirely;

3. the sequential, per-beam lidar ray-march is replaced by a vectorized fixed-length march with a first-hit reduction;
4. the Python-loop collision and proximity tests are replaced by a batched gather of vehicle body points against the corridor;
5. the scalar termination, stop-signal, and spawn branches are replaced by boolean masks with internal automatic reset, so that finished environments restart without host-side control flow.

The batched environment also produces the bird's-eye-view image observation without invoking the Pygame renderer. Because a top-down BEV image is a membership query at a grid of sample points (whether each point lies inside the lane corridor, on an obstacle, or outside the world), it is generated using the same analytic corridor primitives as the lidar march: for each environment an $S \times S$ grid is laid out in the vehicle frame, transformed to world coordinates by the anchor pose, and evaluated at all $N \cdot S \cdot S$ points in a single vectorized gather. The resulting observation is an occupancy image (drivable versus blocked) rather than a rendered grayscale scene, and its cost per environment is comparable to a single lidar frame. All three observation modalities (state, lidar, and BEV image) therefore run on the GPU, so the entire ablation matrix can be executed on the batched environment.

4.9.2 Numerical Parity

Retaining the scalar environment as an oracle allows the batched implementation to be validated for numerical equivalence rather than merely for plausible behaviour. Across a suite of parity tests, the batched dynamics, geometry, observation, and reward pipeline reproduces the scalar model to within floating-point tolerance: the matrix-exponential discretization matches `scipy` to roughly 10^{-8} absolute error, the vehicle dynamics agree to within $\sim 10^{-14}$ over a 200-step rollout, the state and lidar observation vector is bit-identical, and the reward matches to $\sim 10^{-14}$ wherever the proximity term is inactive. The NumPy and CuPy backends produce identical results, confirming that the GPU port introduces no numerical divergence of its own.

Two terms are deliberately not bit-identical, because they replace a rasterized grid with analytic geometry: the near-boundary proximity penalty differs by 0.1–0.5 only within the narrow lane-edge region, and the analytic lidar march agrees with the grid raycast to within approximately one grid cell, with collision decisions matching the grid on 97–98.5% of randomly sampled states (the remainder lying within one cell of the boundary). These differences are confined to the immediate vicinity of boundaries and do not affect the terminal outcomes that drive reward and evaluation, so the batched environment is a faithful stand-in for the scalar reference in every experiment reported here.

4.9.3 Throughput and Scaling

The batched environment was benchmarked on an NVIDIA RTX 4080 SUPER GPU. Table 4.2 reports raw environment-stepping throughput: at large batch sizes the GPU environment sustains roughly 1.7 million transitions per second for the state observation, against about 2,100 for the single-core scalar environment, and roughly 64,000 transitions per second for the compute-heavier 24-beam lidar observation, against about 470. These figures are governed by the batch size and observation type rather than by the vehicle parameters, so they are unaffected by the specific plant used in a given experiment.

Table 4.2: Environment-stepping throughput (transitions per second) for the batched GPU environment versus the single-core scalar reference, measured on an RTX 4080 SUPER.

Observation	Batched GPU (best N)	Scalar (single core)
State (no lidar)	$\sim$1,700,000	$\sim$2,100
Lidar (24 beams)	$\sim$64,000	$\sim$470

Because the identical code runs on either backend, the GPU's contribution can be isolated by swapping only the array library. Table 4.3 shows this crossover: at small N the GPU's per-kernel launch overhead loses to CPU NumPy, but as N grows the GPU pulls ahead, reaching 6.6× at $N = 4096$ for the state observation. For the compute-bound lidar observation the GPU wins at every batch size, up to 62× at $N = 4096$, because each step performs far more work per environment. This crossover is the expected profile of a batched accelerator and motivates the use of large batch sizes ($N \geq 1024$) together with the lidar observation to exploit the device.

Table 4.3: Same batched code, NumPy (CPU) versus CuPy (GPU), by batch size N and observation type. Throughput in transitions per second. Speedup is CuPy relative to NumPy.

Observation	N	NumPy (CPU)	CuPy (GPU)	Speedup
State	256	87,060	30,821	0.35×
State	1024	81,668	122,464	1.5×
State	4096	72,200	479,141	6.6×
Lidar-24	256	2,676	24,815	9.3×
Lidar-24	1024	2,334	71,394	30.6×
Lidar-24	4096	1,005	62,601	62.3×

These gains apply to environment stepping, not to the entire training loop. For off-policy algorithms such as TD3, the gradient-update loop remains the bottleneck: it is a sequence of comparatively small-batch optimizer steps that leaves the GPU underutilized (around 34% utilization during off-policy reverse training), so a single off-policy run is not accelerated in proportion to the environment speedup. The batched environment's value for off-policy learning is therefore twofold: it allows many configuration and seed jobs to share one GPU concurrently, and it makes on-policy algorithms, which consume large environment batches directly and so exploit the fast environment, practical to include in the algorithm comparison of chapter 5. To keep that comparison fair, the batched environment is exposed to Stable Baselines3 [72] through a vectorized-environment adapter, so that the vetted implementations of TD3, SAC, PPO, and DQN run on it unchanged. A separate hand-written GPU-resident TD3 is used only as a throughput demonstration. This infrastructure is what makes the multi-seed ablation campaign of chapter 5 feasible within a practical time budget.

Chapter 5

Results and Discussion

This chapter reports what the learning framework of chapter 4 actually produced. It follows the same path the design took. The algorithm and the reward are settled first on the engineered state, where learning is fast and the outcome is unambiguous. The observation is then opened up, first to lidar and then to the raw image, and the study turns to what richer sensing buys and what it costs. Making the image usable at all turns out to be a training problem rather than a perception problem, and isolating that problem is the central result of the chapter. The failure-replay curriculum, the obstacle-navigation task, and the comparison against tuned classical controllers close it out. Throughout, the aim is to separate two questions that are easy to conflate, whether the articulated control task benefits from richer sensing, and whether any benefit comes from the sensing itself or from the machinery used to turn it into features.

5.1 Evaluation Setup

Every number in this chapter comes from the common evaluation protocol described once in section 4.8, so only the points that bear on reading the tables are repeated here. Policies are scored on a fixed pool of pre-generated scenarios rather than on freshly sampled episodes, so that different controllers face identical layouts. Each configuration is trained over three random seeds and reported as a mean with a 95% confidence interval, so that the stability of a result, and not only its central value, is visible. This matters here because articulated control training is high-variance, and on the harder tasks the spread across seeds carries as much of the story as the mean does. The lane-following results use best-checkpoint completion with the stop action disabled, so that completion measures tracking quality rather than a policy's willingness to decline a layout. The obstacle results, where declining to attempt is itself part of the behaviour, use the attempt-conditional metrics of section 4.8 instead.

Generalization is treated as something to measure rather than assume. The evaluation pool is held out from training, and where two methods are close in mean performance the across-seed interval, rather than the point estimate alone, decides whether the difference is real. A gap of a few centimetres in mean cross-track error routinely sits inside the run-to-run scatter on the reverse task, and reporting a single best number would read that scatter as an effect.

One limitation of the evaluation pool used for the results reported here has to be stated up front. The pool was built by keeping only paths a tuned pure-pursuit controller could complete, which was meant to remove spawn-infeasible layouts that no policy could recover. That filter has a side effect that matters for any comparison against pure pursuit, because it guarantees pure pursuit completes the pool by construction. The pool is therefore being rebuilt with a feasibility criterion that does not privilege any single controller, and every completion and crash rate in this chapter is provisional until that fair-pool re-evaluation lands. The qualitative findings, the orderings, the negative results, and the perception analysis do not depend on the pool. Only the headline rates do.

5.2 Selecting the Algorithm and the Reward

The first question is which learning algorithm and which reward give stable articulated tracking, and the cleanest place to answer it is the engineered state, where perception adds nothing to confound the comparison. Four algorithms, TD3, PPO, a discrete-action PPO, and DQN, were each trained under four reward formulations, in both the forward and the reverse direction, over three seeds. The environment, the observation, and the evaluation were held fixed, and each algorithm was given the settings appropriate to its own update rule as described in section 4.4. Tables 5.1 and 5.2 report best-checkpoint completion for every cell.

Table 5.1: Forward lane-following, best-checkpoint completion rate (%) for every algorithm and reward combination on the engineered state observation. Cells are the mean over three seeds with a 95% confidence interval. The chosen recipe is highlighted.

Algorithm	Dense	Tractor-focus	Multiplicative	Guided (PP)
TD3	100.0±0.0	100.0±0.0	100.0±0.0	100.0±0.0
PPO	100.0±0.0	100.0±0.0	100.0±0.0	100.0±0.0
DQN	100.0±0.0	100.0±0.0	81.5±19.2	57.1±57.5
PPO (discrete)	80.7±37.8	89.0±21.6	100.0±0.0	42.3±3.7

Table 5.2: Reverse lane-following, best-checkpoint completion rate (%) for every algorithm and reward combination on the engineered state observation. Cells are the mean over three seeds with a 95% confidence interval. The chosen recipe is highlighted.

Algorithm	Dense	No-hitch	Multiplicative	Guided (PP)
TD3	81.0$\pm$5.8	83.0$\pm$14.4	**96.9$\pm$1.1**	87.5$\pm$6.0
PPO	14.3$\pm$27.8	13.5$\pm$26.3	43.4$\pm$1.0	81.4$\pm$28.3
DQN	14.7$\pm$28.6	2.0$\pm$3.8	0.1$\pm$0.1	46.4$\pm$55.7
PPO (discrete)	42.3$\pm$2.4	28.9$\pm$28.3	14.5$\pm$28.4	53.6$\pm$8.1

Forward lane-following is saturated. Both continuous-control algorithms complete every forward layout under every reward, which matches the classical baselines, where pure pursuit, PID, LQR, and MPC all reach full completion forward. Only the combination of a discrete action space and a poorly matched reward breaks the forward task at all. Forward tracking is therefore not where the algorithm or the reward is decided, and the forward reward choice is left to the tracking-quality detail rather than to completion.

Reverse lane-following is where the choice is made. The open-loop reverse dynamics are unstable, as established in chapter 3, so early policies fail often and the algorithm has to recover a stabilizing controller rather than merely refine a tracker. TD3 is the only algorithm that solves the reverse task, with every one of its reward variants completing at least 81% of the pool. The best non-TD3 reverse cell, PPO under the guided reward at 81.4%, carries a confidence interval of $\pm$28.3, so its apparent success rests on a single lucky seed rather than on reliable learning. The reward winner is also algorithm-conditional. TD3 pairs best with the multiplicative reward, which forces the tracking and articulation errors down jointly, whereas the on-policy PPO only works under the guided reward that clones the pure-pursuit teacher, and the multiplicative reward is close to catastrophic for DQN. The guided reward fails both discrete algorithms in both directions, which is the expected behaviour of a reward that clones a continuous teacher into an action space that cannot represent it, and serves as a clean negative control.

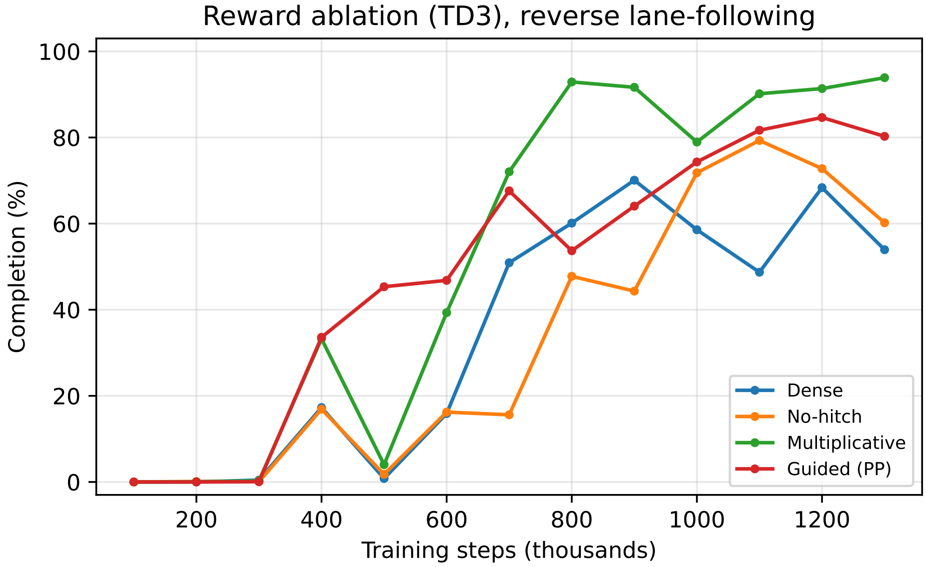


Figure 5.1: Reverse lane-following completion against training steps for TD3 under each reward formulation. The multiplicative reward reaches the highest and steadiest completion.

The chosen recipe is therefore TD3 with the multiplicative reward, and the reverse task is where that choice earns its place. Table 5.3 looks past completion to the tracking detail for TD3, and the multiplicative reward is best on every axis at once, the highest completion at 96.9%, the lowest trailer cross-track error, and the tightest hitch behaviour. This single configuration, TD3 with the multiplicative reward on the engineered state, is the champion carried into the rest of the chapter. Its reverse completion of 0.969 ± 0.011 sits about three points under the pure-pursuit oracle, which clears the safe pool completely, and it does so with no dead seeds. Holding this recipe fixed is what lets every later ablation change one design variable at a time.

Table 5.3: Reverse lane-following, TD3 reward detail on the engineered state observation. Beyond completion the multiplicative reward also gives the lowest trailer cross-track error and the tightest hitch behaviour, which is why it is carried forward. Mean over three seeds, 95% confidence interval on completion.

Reward	Completion (%)	Trailer CTE (m)	Max $\lvert\gamma\rvert$ (°)	Jackknife (%)
Dense	$81.0_{\pm 5.8}$	1.11	17.8	1.8
No-hitch	$83.0_{\pm 14.4}$	0.97	14.2	0.4
Multiplicative	$\mathbf{96.9_{\pm 1.1}}$	0.90	12.0	0.3
Guided (PP)	$87.5_{\pm 6.0}$	1.19	11.5	2.5

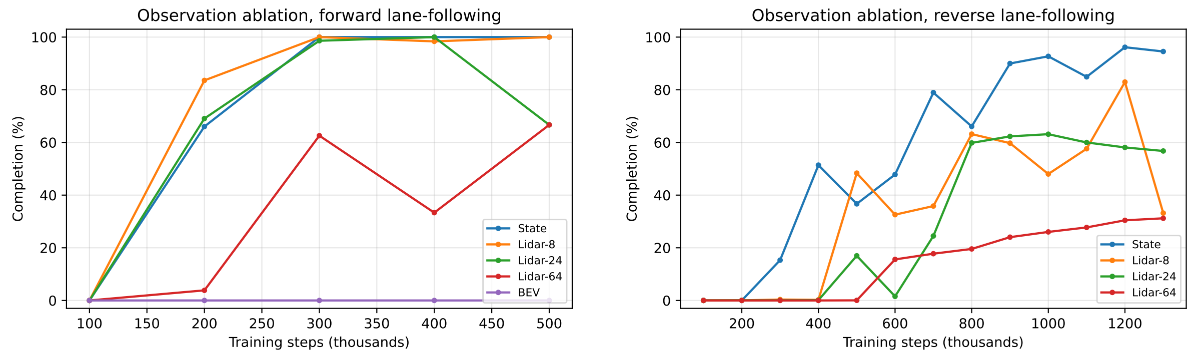
5.3 How Much Sensing the Task Needs

With the algorithm and reward fixed, the observation can be opened up. The engineered state is a compact, hand-designed summary of exactly the quantities a tracking controller needs. Lidar and the bird's-eye-view image carry more of the raw scene but leave the policy to extract those quantities for itself. The question this section answers is what that trade buys. The comparison uses the identical TD3 recipe for every observation, with no curriculum, no failure replay, and no representation warm-up, so that any difference is attributable to the observation and its dimensionality rather than to the training treatment. Table 5.4 reports completion and trailer cross-track error for the engineered state, three lidar resolutions, and the raw image, in both directions.

Table 5.4: Observation-space ablation on no-obstacle lane-following, identical TD3 recipe (multiplicative reward, fixed speed) for every observation. Best-checkpoint completion and trailer cross-track error, mean over three seeds with a 95% confidence interval on completion. Reliability falls as the observation grows, and the effect is sharper in reverse. The reverse BEV run is deferred to the vision-rescue study.

Observation (dim)	Forward completion (%)	Forward CTE (m)	Reverse completion (%)	Reverse CTE (m)
State (8)	$100.0_{\pm 0.0}$	0.34	$97.0_{\pm 0.4}$	1.01
Lidar-8 (10)	$100.0_{\pm 0.0}$	0.41	$90.0_{\pm 12.5}$	1.31
Lidar-24 (26)	$100.0_{\pm 0.0}$	0.46	$63.1_{\pm 61.9}$	1.21
Lidar-64 (66)	$66.7_{\pm 65.3}$	0.95	$31.2_{\pm 61.1}$	1.00
BEV (32×32)	$0.0_{\pm 0.0}$	1.17	—	—

The result runs against the intuition that more sensing is better. Reliability falls as the observation grows, and the effect sharpens on the harder reverse task. The engineered state is not merely sufficient but the most robust option in both directions. The lidar variants show the same pattern internally, which reframes the beam count as a resolution-against-reliability knob rather than a more-is-better axis. The 8-beam scan is the most robust of the lidar options, matching the state forward and reaching 90% reverse, while raising the count to 24 and then 64 beams widens the across-seed spread and then brings on outright seed collapses. The wide confidence intervals on the reverse lidar-24 and lidar-64 rows are not noise to be averaged away, they are the signature of that collapse, with individual seeds falling to zero completion while others still succeed. Because carrying a sensing modality onto hardware is a separate concern taken up in chapter 6, the point retained here is only that added input dimensions cost reliability on this task rather than earning it.



(a) Forward

(b) Reverse

Figure 5.2: Observation-space ablation learning curves, completion against training steps for the identical TD3 recipe under each observation. The raw image never leaves zero, and the higher-dimensional lidar scans learn less reliably than the compact state.

The raw image is the extreme case. Trained from scratch with the same recipe that solves the state task, the bird's-eye-view policy never completes a single forward layout, and the reverse image run was set aside for the dedicated study of the next section. This is a striking outcome, because the image plainly contains the geometry the state encodes, and more besides. Whatever is stopping the image policy from learning, it is not a shortage of information in the observation. That is the puzzle the next section takes apart.

5.4 Learning to Drive from the Image

An image observation that carries strictly more than the engineered state, yet cannot learn the task the state solves easily, is a contradiction worth resolving carefully, because the obstacle task of section 5.6 needs the image. The hypothesis set out in section 4.6 is that the failure is an optimization pathology of off-policy actor-critic learning rather than a limit of perception. A randomly initialized image encoder gives the actor and its convolutional front-end far more capacity to chase a cold, over-optimistic critic into off-distribution actions than the compact state does, and the policy collapses into a degenerate short-horizon behaviour before the critic corrects. This section reports the evidence that isolates that cause and then the recipe that removes it.

5.4.1 Isolating the Cause

The evidence takes the form of a ladder, where each rung changes exactly one thing relative to the rung above, so that the first configuration to recover completion identifies what was actually broken. Table 5.5 lays it out on the forward task.

Table 5.5: The vision-training ladder on forward lane-following. Each rung changes one thing relative to the rung above, so the first configuration that reaches full completion identifies the cause. Representation changes alone leave the policy at zero, imitation restores it, unconstrained fine-tuning destroys it again, and only freezing the actor holds it, which points to the actor rather than the encoder as the failure. Completions are from the development runs. The actor-frozen configuration was repeated over three seeds (1.00 / 1.00 / 0.92), and per-seed archival of the remaining rungs is pending.

Configuration	Completion	What it isolates
BEV occupancy, from scratch	0.00	Ordinary RL on the raw image fails outright
+ signed-distance encoding	0.00	A denser, task-aligned image is not the fix
+ coordinate channels	0.00	Spatial addressing is not the fix
+ stability extractor	0.00	Layer normalization and separate actor and critic encoders are not enough
Imitation warm-start	1.00	A cloned state expert produces a working vision policy
+ naive fine-tuning	0.00	Letting RL move the whole policy destroys the clone
+ RL, encoder frozen	0.00	Freezing perception does not save it, so the encoder is not the culprit
+ RL, actor frozen	1.00	Freezing the actor and training only the critic holds parity

The representation-side fixes come first and none of them works. Rendering the image as a graded signed-distance field rather than a hard occupancy mask, adding coordinate channels, and giving the encoder the stabilizing treatment of layer normalization with separate actor and critic front-ends all leave completion at zero. If the trouble were that the image is a poor learning target, one of these would have moved the needle. None does, which points away from the representation as the cause. The picture changes the moment the policy is warm-started by imitation. Cloning the competent state expert into the image policy produces a working controller with full completion, which shows that a non-degenerate policy over the image is reachable. The decisive rungs are what happens next. Letting ordinary reinforcement learning then fine-tune that cloned policy destroys it, dropping completion back to zero, and freezing the encoder while the rest of the policy trains does not save it either. Only freezing the actor, and training the critic alone, holds the cloned policy at parity. Read down the ladder, the culprit is neither the image nor the encoder but the actor being dragged off-distribution by an untrained critic, exactly the mechanism hypothesized in section 4.6. The actor-frozen configuration was repeated over three seeds and held, at 1.00, 1.00, and 0.92 completion.

This diagnosis also exposes why the imitation warm-start, though it works, is not a satisfying end point. Because the image observation still carries the full state vector, the cloning objective can be met largely through the state branch, so a working cloned policy does not on its own prove that the convolutional encoder learned to read the image. Settling that requires an approach where the image has to carry its weight.

5.4.2 Distilling Geometry into a Frozen Encoder

The recipe that removes the failure at its root refuses to let reinforcement learning shape the encoder at all, following section 4.6. A convolutional encoder is trained by supervised regression to predict the vehicle's geometry directly from the bird's-eye-view image, using the simulator's true state as a privileged teacher label. It is then frozen and used as a fixed perception module, and the policy is trained from scratch with ordinary TD3 over a low-dimensional vector, exactly as in the engineered-state baseline. The internal steering and hitch angles are supplied as true proprioception, since they are cheap direct measurements and are not reliably visible in a top-down image, while the exteroceptive path geometry is read from the frozen encoder.

Two measurements test this recipe. The first asks whether the image is a sufficient statistic for the geometry, by scoring how well each state component can be predicted from the image alone. Table 5.6 reports the per-component test R^2. The exteroceptive geometry, the tractor and trailer tracking errors and the two curvature look-aheads, is recovered almost perfectly, with R^2 at or above 0.97 for seven of the eight components. Only the internal steering angle resists, at 0.66, which is exactly the component that a top-down view of the surroundings should not be able to see, and precisely why steering is supplied as proprioception rather than predicted. Because predicting the state from the image alone cannot be satisfied through a state branch, this is the direct evidence the imitation route could not provide, the image really does encode the path geometry.

Table 5.6: Per-component test R^2 of predicting each state variable from the bird's-eye-view image alone, using the frozen geometry encoder. The exteroceptive path geometry is recovered almost perfectly, while the internal steering angle resists, which is why steering and the hitch angle are supplied as true proprioception rather than predicted.

Component	Description	R^2
e_ψ	tractor heading error, exteroceptive	0.998
e_y	tractor cross-track error, exteroceptive	0.997
κ_1	near curvature sample, exteroceptive	0.997
κ_2	far curvature sample, exteroceptive	0.997
γ	hitch angle, partly visible through the path history	0.995
$e_{\psi,t}$	trailer heading error, exteroceptive	0.993
$e_{y,t}$	trailer cross-track error, exteroceptive	0.974
δ	steering angle, an internal actuator state the image cannot see	0.659

The second measurement asks whether a policy can then learn on these frozen features. Figure 5.3 shows the from-scratch learning curve. Completion climbs monotonically to full parity by roughly 300 thousand steps, with no imitation, no critic warm-up, no unfreezing schedule, and no collapse, tracing the same trajectory as the engineered-state run. The image policy that could not learn at all in section 5.3 learns cleanly once the encoder is decoupled from the reward-driven gradient. Cross-track error at parity is looser than the state expert's, on the order of [pending: geometry-encoder CTE at parity], which is consistent with the few-percent prediction error the encoder carries.

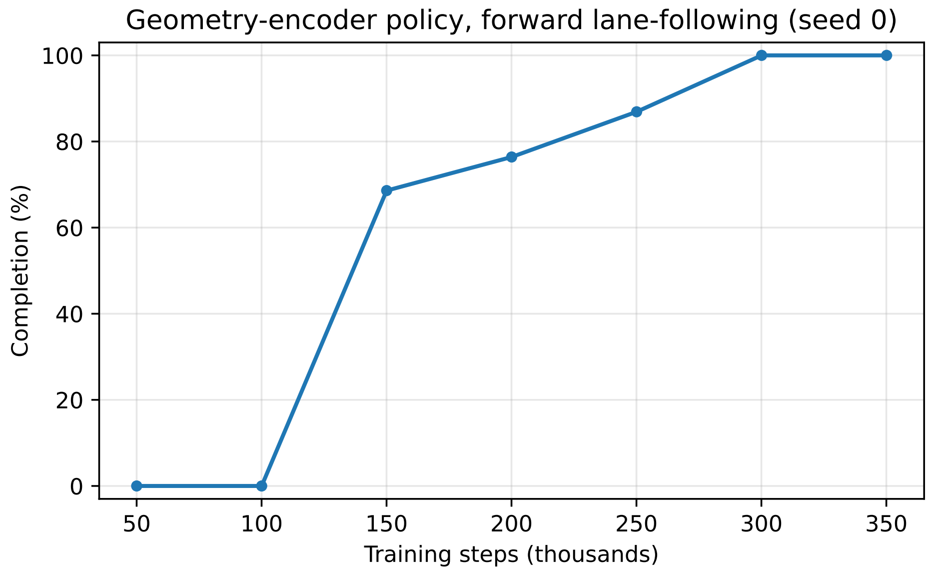


Figure 5.3: Geometry-encoder policy completion against training steps on forward lane-following, trained from scratch on the frozen encoder. Completion reaches parity with the engineered-state baseline by about 300 thousand steps with no collapse.

Two limits are worth stating plainly. The clean curve is a single seed, and the multi-seed confirmation together with the reverse direction are [pending: geometry-encoder three-seed and reverse results], held up by a stochastic native crash under the per-step encoder workload rather than by anything in the method. This is a gap in statistical rigour, not in validity, since the recipe reached full parity. Separately, the question of whether reinforcement learning can safely improve a warm-started policy without re-triggering the collapse was probed directly. Freezing the actor holds parity but by construction never improves the policy, and allowing the actor to move again, even under a reduced learning rate, either collapsed the policy outright or left it oscillating with no improvement in cross-track error, [pending: safe-adaptation collapse rates and CTE]. Unconstrained actor unfreezing is therefore not yet a usable recipe on this task, and whether the lane-tracking reward even leaves headroom for adaptation to capture is left open.

5.5 Recycling Failures during Training

The failure-replay curriculum of section 4.7 is the one training treatment expected to help most exactly where the task is hardest. It reuses the seed of any failed episode so the agent meets the same hard layout again rather than discarding it for a fresh random one, and it is evaluated as a controlled ablation against a random-reset baseline that is identical in every other respect. The reverse task is the natural place to look for the effect, because the reverse failure rate is high early in training and it is those early failures that the curriculum recycles into extra learning signal. For forward lane-following the failure rate falls to near zero after minimal training, so little benefit is expected there.

The comparison is read from the learning curve and from the final policy together, since a curriculum may sharpen sample efficiency without raising the asymptotic ceiling. Table 5.7 sets out both, the steps to a fixed completion level as the efficiency measure and the held-out tracking and stability metrics as the final-quality measure. [pending: failure-replay ablation numbers] are awaiting import from the paired runs.

Table 5.7: Failure-replay curriculum against a random-reset baseline on reverse lane-following, identical observation, reward, and hyperparameters. Steps to a fixed completion level measure sample efficiency, the remaining columns measure the final policy. [pending: numbers awaiting import from the failure-replay runs.]

Reset strategy	Steps to 50% completion ($\times 10^3$)	Trailer CTE (m)	Max $\lvert\gamma\rvert$ (°)	Completion (%)	Jackknife (%)
Random reset	—	—	—	—	—
Failure replay	—	—	—	—	—

Figure unavailable

Reverse lane-following learning curves for the failure-replay curriculum against a random-reset baseline, identical hyperparameters, observation, and reward.

`figures/experiment_results/failure_replay_curve.pdf`

Figure 5.4: Reverse lane-following learning curves for the failure-replay curriculum against a random-reset baseline. Both agents share hyperparameters, observation, and reward, so the curriculum is the only difference.

5.6 Navigating Around Obstacles

The obstacle task is where the whole argument of the chapter comes together. It is the one setting where the engineered state is genuinely insufficient, because a compact tracking summary is blind to obstacles that are not on the reference path, so the image has to carry information the state cannot. It is also where the value of a learned controller separates cleanly from classical tracking, because avoiding an obstacle is a joint planning-and-control problem rather than a pure tracking one. The approach taken here is not a monolithic mapping from pixels to actions, which section 5.4 showed does not train. It is a learned but interpretable geometric bottleneck, and the results below build it up one decision at a time.

5.6.1 Encoding Obstacles Directly Does Not Work

The direct approach is to hand the policy the obstacles, as a set of ordered range-and-bearing slots appended to the state. It fails. Trained under the same curriculum, the slot-based policy climbs to a modest completion and then collapses toward zero, and it does so under every slot ordering tried, nearest-first, fixed, and randomized alike. Because a fixed ordering with no reordering collapses just as the others do, the instability is not caused by slots swapping identity between steps. It is the same off-policy training pathology seen in section 5.4, now provoked by a different high-dimensional input. This rules out explicit obstacle encoding and motivates a representation that does not ask the policy to parse a variable set of objects at all.

5.6.2 Observing the Hidden Avoidance Path

The idea that works is to give the policy a path rather than a scene. The environment already computes a hidden avoidance path, a potential-field offset from the centreline that bends around the obstacles, and the reward already tracks it. Instead of feeding the policy the obstacles, the policy is fed the tracking errors and curvature to this avoidance path. Obstacle avoidance then reduces to path following, the task already solved in the preceding sections, and the obstacle information arrives pre-digested as the path to follow. The image predicts this path, in the same learning-by-cheating manner as the geometry encoder of section 5.4.2, privileged in simulation and image-derived at deployment.

A ground-truth test isolates the mechanism from the perception. Taking an existing forward state policy and feeding it avoidance-path errors in place of centreline errors, with no retraining at all, cut obstacle collisions roughly in half and lifted completion from about 0.55 to about 0.67. The controller follows whatever path it is given, so the entire obstacle problem can be moved into the choice of path, which is exactly the quantity the image can supply.

5.6.3 Deciding When to Stop, and the Forward Result

Following the avoidance path is not enough on its own, because some layouts are not passable and the right action there is to stop rather than to clip an obstacle. The policy is therefore given a windowed difficulty channel, the free-gap difficulty of eq. (4.19) computed over the obstacles inside the observable window, so that the attempt-versus-stop decision is a causal function of what the agent can currently see. As set out in section 4.3.1, keying the stop reward on this same windowed difficulty is what makes the value of a stop predictable from the observation. An earlier design that keyed the reward on a global per-episode difficulty made the stop value unpredictable from the observation and collapsed the policy into stopping everywhere, and sharpening the stop calibration, rather than lowering its magnitude, concentrated stopping on the genuinely blocked layouts.

In the forward direction this static-difficulty recipe works well. Table 5.8 reports the outcome stratified by layout difficulty. The policy drives through easy layouts, stops on every hard one, and does not crash on the hardest, for an overall safe rate of 0.907 at a 0.019 crash rate. The residual crash concentrates in the tight-but-passable band where the attempt-versus-stop call is genuinely ambiguous. This behaviour is what the drivability calibration of section 4.8 predicts. Calibrating the geometric difficulty against the classical drivability oracle through the expected-value balance of eq. (4.27) puts the forward break-even near a difficulty of 0.8, so stopping on the hardest layouts is the rational choice rather than a shortfall.

Table 5.8: Forward obstacle manoeuvring, retained static-difficulty model, outcomes stratified by layout difficulty. The policy drives through easy layouts and stops on every hard one with no crash on the hardest, for an overall safe rate of 0.907 at a 0.019 crash rate. The residual crash concentrates in the tight-but-passable band. Rates are provisional pending the fair-pool re-evaluation.

Difficulty	Complete (%)	Stop (%)	Crash (%)
$[0.2, 0.4)$ easy	83.1	15.5	1.4
$[0.4, 0.6)$	53.0	40.8	6.3
$[0.6, 0.8)$	0.0	100.0	0.0
$[0.8, 1.0)$ blocked	0.0	100.0	0.0

5.6.4 The Reverse Case Needs a State-Dependent Difficulty

Reverse is the culmination, and the forward recipe does not transfer to it. A static difficulty describes the layout, the free gap, but not the situation, and in reverse the two come apart. Because reverse tracking runs a systematic off-path error of around 0.8 m, tracking well at one moment does not imply the trailer will clear the coming gap, so a policy gated on layout difficulty over-attempts in the ambiguous band and clips obstacles there, crashing on around forty to forty-five percent of the attempts it makes in that band.

The fix is to make the difficulty state-dependent. The effective vehicle width is inflated by the current trailer error to the avoidance path rather than by an average, so when the trailer is tracking off-path into an obstacle the usable gap shrinks now and the difficulty rises now, in time to trigger a stop. A second correction concerns where the difficulty looks. In reverse the trailer leads, so a window that looked only ahead of the lead point was blind to obstacles the trailing body was still clearing, which produced a spurious cluster of crashes at apparently low difficulty. Anchoring the window to the whole vehicle body removed it. This is a reverse-specific need, as the collision geometry in table 5.9 makes plain. Reverse crashes are overwhelmingly trailer clips, the leading trailer sweeping into an obstacle, whereas forward crashes are entirely tractor clips, so the body-extent window matters in reverse and is irrelevant forward.

Table 5.9: Collision geometry by direction. Reverse crashes are dominated by trailer clips, which the body-extent difficulty window addresses, while forward crashes are entirely tractor clips, so the window fix is a reverse-only need.

Direction	Tractor clip (fraction)	Trailer clip (fraction)
Forward	1.00	0.00
Reverse (mean of 3 seeds)	0.18	0.82

With the state-dependent difficulty and the body-extent window in place, the stop decision reduces to a single clean threshold. Measured at the moment of collision, no reverse crash occurs below a dynamic difficulty of about 0.5, so a stop below that value is unambiguously wrong and a stop at or above it is the best available action. Figure 5.5 shows the outcome stratified by dynamic difficulty, and table 5.10 gives the same data numerically. Completion falls and stopping rises across the gate, and the crash rate stays low throughout, with the accepted model reaching an overall crash rate of about 0.05.

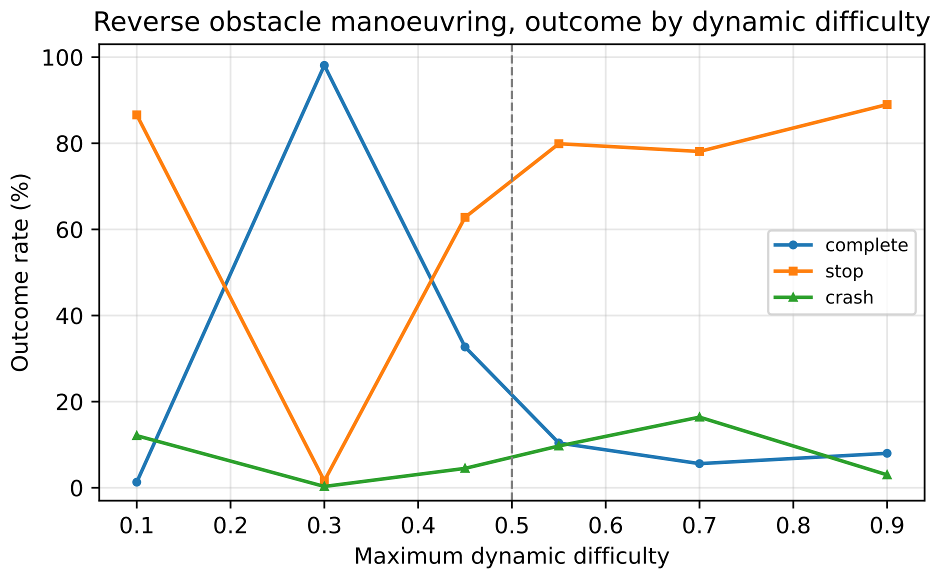


Figure 5.5: Reverse obstacle manoeuvring, outcome rates against maximum dynamic difficulty. No crash occurs below the gate near a dynamic difficulty of 0.5, so stopping above it is the correct action.

Table 5.10: Reverse obstacle manoeuvring, accepted dynamic-difficulty model, outcomes stratified by the maximum dynamic difficulty reached. Completion falls and stopping rises with difficulty, and no crash occurs below the gate at roughly 0.5. The low-difficulty stops are the accepted articulation limitation, not spurious behaviour. Rates are provisional pending the fair-pool re-evaluation.

Dynamic difficulty	n	Complete (%)	Stop (%)	Crash (%)	Stop $\|\gamma\|$ (rad)
$[0.0, 0.2)$	762	1.3	86.6	12.1	0.204
$[0.2, 0.4)$	2239	98.1	1.6	0.3	0.105
$[0.4, 0.5)$	584	32.7	62.8	4.5	0.039
$[0.5, 0.6)$	1160	10.4	79.9	9.7	0.101
$[0.6, 0.8)$	1217	5.6	78.1	16.4	0.119
$[0.8, 1.0)$	5778	8.0	89.0	3.0	0.062

The reverse recipe is trained from scratch and repeated across three seeds in table 5.11. Two of the three reproduce the safe result at a crash rate around 0.05, while one seed converged to a less conservative policy that attempts more of the transition band and crashes more there. The spread traces to a critic shock at the boundary where the reward changes from the obstacle-free warm-up to the obstacle task, so the recipe is usable but not yet tightly reproducible. A critic warm-up at that boundary and a safety-aware checkpoint criterion are the levers to tighten it, and both are left as immediate follow-ups. The same table records a second finding. Dynamic difficulty helps reverse but hurts forward, where the vehicle does not jackknife and its tracking error is small, so the width inflation only adds noise. Forward therefore keeps the static recipe, and the hope of a single unified stop signal for both directions does not survive contact with their different dynamics.

Table 5.11: Reverse dynamic-difficulty model across three seeds trained from scratch. Two of three seeds reproduce the safe result, and seed 1 converged to a less conservative policy. Rates are provisional pending the fair-pool re-evaluation.

Seed	Complete (%)	Stop (%)	Crash (%)
Seed 0	26.0	68.8	5.2
Seed 1	36.8	49.7	13.5
Seed 2	32.8	62.3	5.0
Mean ± sd			$7.9_{\pm 4.9}$

Table 5.12: Forward crash rate, static against dynamic difficulty. Dynamic difficulty hurts forward, so forward keeps the static recipe. Rates are provisional pending the fair-pool re-evaluation.

Forward recipe	Crash (%)
Static difficulty	2.7
Dynamic difficulty (mean of 3 seeds)	8.7

5.6.5 Reading the Difficulty from the Image

The obstacle recipe gates on the dynamic difficulty, which depends on the current trailer error and so is a harder image target than the static path geometry of section 5.4.2. The same distillation recovers it. Table 5.13 reports the per-component recoverability of the avoidance-path state from the obstacle image in both directions, and the dynamic difficulty channel is recovered well, at a test R^2 of 0.948 forward and 0.954 reverse. The avoidance-path errors are recovered strongly, while the avoidance-path curvature is softer than in the lane task and softer still in reverse, where the bends are sharper, though the errors and the difficulty channel carry the control.

Table 5.13: Recoverability of the avoidance-path state from the obstacle image, test R^2 per component for the forward static and reverse dynamic recipes. The dynamic difficulty channel is highly recoverable in both directions, so there is no perception roadblock. Reverse avoidance-path curvature is softer than forward.

Component	Forward R^2	Reverse R^2
e_y	0.947	0.967
e_ψ	0.870	0.876
$e_{y,t}$	0.977	0.977
$e_{\psi,t}$	0.985	0.882
κ_1	0.699	0.425
κ_2	0.558	0.277
d_{win}	0.948	0.954

Closing the loop confirms there is no perception roadblock. Table 5.14 drives each policy on its own image-predicted state, with a frozen encoder in the loop and only the steering and hitch angles taken from proprioception. The full pipeline holds in both directions, with the crash rate sitting a few points above the ground-truth-state policy from ordinary prediction noise, a larger margin in reverse where the curvature is harder to see. The image supplies both the plan, the avoidance path and its difficulty, and the control state, which is the claim the whole perception thread of this chapter was built to support.

Table 5.14: Closed-loop vision pipeline, policy driven on image-predicted avoidance-path state with a frozen encoder in the loop. The full pixels-to-control pipeline holds in both directions, a few points of crash above the ground-truth-state policy from prediction noise. Rates are provisional pending the fair-pool re-evaluation.

Recipe	Difficulty R^2	SAFE (%)	Crash (%)
Forward (static)	0.948	90.3	4.1
Reverse (dynamic)	0.954	74.9	8.3

5.6.6 An Accepted Limitation

One behaviour is worth naming as a limitation rather than hiding. On predominantly straight reverse paths the agent occasionally halts where a continued manoeuvre would have completed. This is not spurious stopping. At these halts the trailer's hitch articulation is measurably degrading, with a terminal articulation angle around 0.20 rad against about 0.01 rad on completed paths, trending toward, though stopping short of, the unrecoverable regime around 0.75 rad that produces the collisions. A reversing tractor-trailer cannot unwind a growing hitch angle by continuing in reverse, because the reverse articulation dynamics are unstable and more reverse motion only grows the hitch. The geometrically required recovery is to pull forward, re-stabilize the articulation on the stable forward dynamics, and re-approach, which is a shunting manoeuvre. That recovery needs a forward-reverse switching action space that is outside the scope of this continuous-reverse study, so halting is the correct and safe reverse-only response. The policy errs conservative, aborting as the articulation begins to degrade rather than at the point of no return, and trades completion throughput for safety. Permitting pull-forward recovery is left to future work.

Taken together, these results answer the question of why a learned controller is worth the trouble here when a classical planner exists. The value is twofold. It is the learned attempt-versus-abort stop decision, which a geometric planner does not make, and it is the reverse articulated case, where a geometric avoidance path is not by itself trackable and the stop must be tied to the articulation state. Both live in one learned module that reads its plan and its control state from the same image, in place of the separate global planner, local planner, and tracker of the classical stack. The comparison against that stack is taken up next.

5.7 Comparison with Classical Controllers

A learned controller has to be shown competent at plain tracking before its obstacle behaviour means anything, so this section places it beside tuned classical control on the no-obstacle task. The framing matters. On pure path tracking a classical controller is complete and verifiable, and for that task it is the sensible choice, so the aim here is not to beat it but to establish that the learned controller reaches comparable tracking competence. The classical controllers were tuned by differential evolution and validated on held-out seeds, and every controller faces the same pool as the learned agent, subject to the fair-pool caveat of section 5.1. Tables 5.15 and 5.16 report the comparison.

Table 5.15: Forward lane-following controller benchmark on the safe pool. Tracking is saturated for every method, so completion cannot separate them here.

Controller	Completion (%)	Trailer CTE (m)	Max $\|\gamma\|$ (°)
TD3 (state, multiplicative)	100.0$\pm$0.0	0.452	8.2
Pure pursuit	100.0	0.146	5.0
PID	100.0	0.167	5.5
LQR	100.0	0.262	5.9
MPC	100.0	0.272	6.0

Table 5.16: Reverse lane-following controller benchmark on the safe pool. The learned policy matches pure pursuit on completion and clears PID, while MPC, which receives an explicit path preview, is the informed upper bound.

Controller	Completion (%)	Trailer CTE (m)	Max $\|\gamma\|$ (°)
TD3 (state, multiplicative)	96.9$\pm$1.1	0.901	12.0
Pure pursuit	100.0	0.841	7.0
PID	89.8	0.788	14.7
LQR	96.7	0.568	9.5
MPC	99.0	0.844	7.8

Forward tracking is saturated, so completion cannot separate the methods and the comparison there is only about tracking quality. Reverse is more informative. The learned policy reaches completion comparable to pure pursuit and ahead of PID, which establishes the competence the obstacle results depend on, a policy that can actually stabilize and track the trailer on the unstable reverse task. This is a competence result rather than a claim of superiority, and the caveat of section 5.1 applies with force. The pool was defined by pure-pursuit success, so pure pursuit completes it by construction, and the honest reverse comparison awaits the fair-pool re-evaluation. Even then the case for the learned controller does not rest on winning this tracking comparison. It rests on the integrated obstacle task of section 5.6, where pure pursuit alone is not a controller and the classical alternative is the full planner-and-tracker stack. In articulated control, success is not simply the lowest cross-track error, because hitch stability takes precedence over exact path following. A policy that holds the hitch angle inside a safe margin throughout a reverse manoeuvre is more successful than one that shaves cross-track error but risks jackknifing, which is why the benchmark reports completion, trailer tracking, and articulation stability together rather than collapsing them into one score.

5.7.1 Information Parity between Learned and Classical Controllers

A controller comparison is only meaningful if the methods are given comparable information about the reference path. Table 5.17 summarizes what each method observes at a single control step on the no-obstacle task. The TD3 observation contains the instantaneous tractor and trailer tracking errors together with two path-curvature samples taken ahead of the trailer axle. This is a strict superset of the reference information used by pure pursuit and PID, which close the loop on the same instantaneous errors plus a single curvature sample at one lookahead point. The learned policy is therefore not at an information disadvantage relative to these baselines, if anything it is marginally favoured, so the comparison against pure pursuit and PID is a fair, like-for-like test of control quality rather than one confounded by what each method is allowed to see.

Table 5.17: Reference-path information available to each method at a single control step on the no-obstacle task. The learned policy's observation is a strict superset of the information used by the pure-pursuit and PID baselines. Only MPC receives an explicit preview of the path ahead.

Method	Instantaneous errors (e_y, e_ψ)	Curvature lookahead	Full path preview
Pure pursuit	yes	1 pt (∼10 m)	no
PID	yes	1 pt (∼10 m)	no
TD3 (state/lidar/BEV)	yes	2 pts (∼10, 20 m)	no
MPC	implicit	no	N=16/20 steps

MPC is the exception, and it is best read as an informed upper bound rather than a peer. It is supplied with an explicit preview of the path, a full reference trajectory over a receding horizon reconstructed from the same centreline the policy tracks, while the learned policy receives only the two scalar curvature samples. Where MPC attains lower cross-track error, part of that advantage is therefore attributable to horizon information the learned policy never sees, rather than to a superior control law alone. The pure-pursuit and PID comparisons are the information-matched test of whether a learned policy can equal classical feedback control on the articulated tracking task, and closing the remaining gap to MPC by giving the policy an equivalent path-preview observation is identified as future work in chapter 7.

For obstacle scenarios the benchmark question changes, because the learned obstacle policy is solving a joint local-planning and control problem, including speed modulation, whereas the classical controllers are path trackers. The obstacle results are therefore not presented as directly comparable to controller-only baselines unless those baselines are given a matched planning layer and longitudinal policy.

5.8 Failure Modes and Deployment Relevance

Average metrics alone are insufficient for articulated-vehicle control, because the operational significance of a failure depends on how it occurs. The analysis therefore examines representative failure cases for each controller family, such as gradual corner-cutting, growing lateral oscillations, abrupt jackknife onset, and obstacle-clearance failures, rather than mean error alone. Trajectory overlays and hitch-angle traces are the natural diagnostics, because they reveal whether an error mode is geometric, dynamic, or perception-driven, and therefore which part of the pipeline a fix should target. The seed collapses behind the wide confidence intervals of section 5.3 are a case in point, since they are not a uniform degradation but a bimodal outcome where a policy either learns the task or falls into a degenerate behaviour, which is visible in a trajectory trace in a way a mean completion rate hides.

This failure-oriented reading is also what connects the simulation study to deployment. A controller that attains a low mean cross-track error but exhibits rare catastrophic jackknife episodes is less deployable than one with slightly higher average error and consistent stability margins, which is why the reverse results are read through hitch stability rather than cross-track error alone. For the same reason controller latency and architectural modularity are weighed alongside tracking performance. Taken together, the results position the learned controller as a credible peer to classical feedback control on the unstable reverse task, and the geometry-encoder recipe as the piece that makes high-dimensional perception trainable rather than merely available, which is what the obstacle task and eventual hardware deployment in chapter 6 require.

Chapter 6

Sim-to-Real Deployment

6.1 Overview

The preceding chapters developed and evaluated an articulated tractor-trailer controller entirely within a 2D simulation environment. This chapter describes the transfer of that controller to a physical robotic platform, constituting the sim-to-real component of this thesis. Unlike a conventional model-based stack, the deployed system does not re-implement a hand-derived tracking law on the vehicle. Instead, the simulation-trained reinforcement-learning (RL) policies are executed directly on hardware through a thin bridging layer that reproduces the observation and action interfaces the policies were trained against. The objective is to demonstrate that the vehicle model, the observation definitions, and the learned control logic established in simulation can be ported to real hardware with minimal algorithmic redesign, and to characterize the engineering challenges encountered during that transfer.

The deployment is built on a deliberately stripped-down Autoware stack. The standard Autoware localization and perception pipelines are retained essentially unchanged, while the behaviour, planning, and control layers are replaced by a custom RL bridge that runs the forward and reverse policies. Two pieces of supporting infrastructure are described in detail because they are specific to the articulated, sensor-light platform: an onboard LiDAR-based hitch-angle estimator that recovers the articulation angle without any trailer-mounted sensor, and a Smith predictor [42] that compensates for actuator delay so that delay-free-trained policies behave correctly on a lagging physical actuator.

6.2 Hardware Platform

6.2.1 Tractor Vehicle

The physical tractor is an Agilex Hunter SE, a compact electric wheeled unmanned ground vehicle with an Ackermann steering geometry. The relevant mechanical parameters are a wheelbase of 0.65 m, a wheel radius of 0.15 m, and a maximum steering angle of 0.436 rad (25°). Vehicle actuation is commanded over a CAN bus on a 20 ms cycle. Speed commands are encoded as 16-bit signed integers in units of 1 mm/s (the commanded speed in m/s scaled by 1000) and steering commands as 16-bit signed integers scaled by approximately 1320 counts per radian (≈ 0.8 mrad per count).

6.2.2 Trailer

The trailer is coupled to the tractor through a revolute hitch joint located marginally behind the tractor rear axle, which was added to the tractor-trailer model used for training. The lab trailer has a wheelbase of approximately 2.8 m and a body width of roughly 0.33 m. The articulation angle γ, defined as the difference between the tractor and trailer headings ($\gamma = \psi_{\text{truck}} - \psi_{\text{trailer}}$), is estimated onboard from the tractor's own LiDAR rather than measured by any trailer-mounted sensor. The procedure is detailed in Section 6.3.4. The estimate is published to the ROS2 network on the `/vehicle/trailer_state` topic. The control bridge consumes the most recent published value, with no staleness gating currently applied. Two distinct angular limits appear in the stack and should not be conflated. The onboard estimator clamps the published articulation estimate to $|\gamma| \leq 90°$, whereas the deployment trailer model defines a tighter operational limit of $|\gamma| \leq 80°$ (1.40 rad), itself below the 90° jackknife condition used to terminate simulation episodes.

6.2.3 Sensor Suite

The vehicle is instrumented with the following sensors:

- **LiDAR**: RoboSense Helios mechanical LiDAR mounted at the front of the tractor, publishing 3D point clouds on `/rslidar_points`. This single sensor is reused for three purposes: NDT-based localization [83], environment perception, and onboard hitch-angle estimation.
- **Cameras**: Six Basler Pylon industrial cameras (rear, rear-right, centre, left, merging, and right) producing compressed colour images. The cameras are not consumed by the LiDAR-based RL policies and are used for logging and monitoring only.
- **IMU**: WitMotion 6-DoF inertial measurement unit, with a gyroscope-bias estimator applied in the localization pipeline.
- **GNSS**: u-blox ZED-F9P receiver with RTK corrections, providing centimetre-level position estimates converted to MGRS coordinates for Autoware localization.
- **Wheel odometry**: Vehicle speed and steering-angle feedback recovered from CAN messages and fused with IMU data for dead-reckoning between GNSS and scan-matching updates.

6.3 Software Architecture

6.3.1 Middleware and Framework

The deployment stack runs on ROS2 Humble (Ubuntu 22.04) [84] and is integrated into the Autoware autonomous-driving framework [85]. Autoware provides modular localization, perception, planning, and control pipelines whose interfaces are defined by standardized ROS2 message types. As introduced in the overview, this deployment keeps the standard sensing, localization, and perception modules and replaces only the planning and control layers with the learned RL bridge.

6.3.2 Relationship to the Conventional Articulated Pipeline

For context, Autoware's conventional approach to articulated vehicles is fully model-based. A truck-trailer mode manager routes goals to a freespace planner (a hybrid A* search [15] that respects the non-holonomic constraints of the articulated vehicle), and the resulting reference trajectory is tracked by a model-predictive or LQR lateral controller. This model-based pipeline exists within the wider project and represents the standard, well-understood baseline for trailer control. The relevant control background is reviewed in Chapter 2. The contribution of this thesis, however, is the learned controller, and the deployment described here therefore routes control through the RL bridge instead of the planner/MPC chain. The two are mutually exclusive at run time and are selected at launch.

6.3.3 RL Bridge Node Architecture

The learned control path comprises the following principal nodes:

`lane_reference_node` Loads the Lanelet2 map, selects the active lane from the ego odometry and the current goal, and publishes the lane centreline as a reference path together with a `drive_enabled` flag and a `drive_direction` flag (forward or reverse). This node supplies the path geometry against which the policy's tracking observation is computed.

`rl_bridge_node` The core of the deployment. It loads the trained forward and reverse TD3 policies (Stable-Baselines3 [72] / PyTorch [86] checkpoints) and runs a 10 Hz control loop. On each tick it assembles the policy observation in exactly the layout used during training, an 8-dimensional state vector (steering angle, articulation angle, tractor and trailer tracking errors, and look-ahead path curvature) concatenated with a 24-beam LiDAR range vector, queries the active policy deterministically, and converts its 2-dimensional action (a steering rate and a stop/go signal) into the steering-angle and speed commands expected by the vehicle interface, with the stop/go signal mapped onto the task's fixed speed or a halt exactly as the simulation interface layer does during training. Direction-dependent post-processing (action smoothing and kinematic rate scaling) is applied for the reverse policy, whose outputs are inherently more oscillatory than the forward policy's.

`hitch_angle_estimator` Estimates the articulation angle from the tractor LiDAR and publishes it on `/vehicle/trailer_state` (Section 6.3.4).

`initialpose_shim` Bridges the RViz `/initialpose` message to the Autoware pose-initializer service, allowing the operator to seed localization interactively.

`controller_canbridge` Translates `autoware_control_msgs/Control` commands into Hunter SE CAN frames and parses vehicle feedback (velocity, steering angle) back into ROS2 topics.

The forward and reverse policies are loaded from separate checkpoints and selected according to the `drive_direction` flag, so that the same bridge handles both manoeuvring regimes without retraining or reconfiguration.

6.3.4 Hitch Angle Estimation

Because the trailer carries no articulation encoder and no trailer-mounted sensing, the articulation angle is recovered geometrically from the tractor's front LiDAR, which observes the front (and, at larger angles, the side) face of the trailer. The estimator operates on each incoming point cloud as follows.

Region of interest. Points are first cropped to a box in the LiDAR frame that brackets the expected location of the trailer face. The lateral extent of this region is shifted to track the face as the trailer swings: using the current filtered estimate γ, the lateral offset is set to

$$\Delta y = -\, D_{\text{face}} \sin \gamma, \tag{6.1}$$

where D_{face} is the approximate longitudinal distance from the hitch pivot to the face. This keeps the cropped points on the trailer face across the full range of articulation rather than only near $\gamma = 0$.

Rectangle fitting. Rather than fitting a single planar normal, the estimator fits the orientation of the trailer body to the cropped points using the minimum-closeness rectangle criterion of Zhang et al. [87]. For a candidate orientation θ, the points are rotated into axis-aligned coordinates (u, v) and scored by their closeness to the nearest edges of the bounding rectangle,

$$C(\theta) = \sum_i \min\big(d_i^u,\, d_i^v\big)^2, \qquad d_i^u = \min\big(|u_i - u_{\min}|,\, |u_i - u_{\max}|\big), \tag{6.2}$$

with d_i^v defined analogously. The fitted orientation minimizes $C(\theta)$ augmented by a dimension-prior penalty that favours rectangles consistent with the known trailer width and length:

$$\theta^\star = \arg\min_{\theta \in [0,\pi)} \big[\, C(\theta) + \lambda\, P_{\dim}(\theta) \,\big]. \tag{6.3}$$

This formulation is robust to whether only the front face, both the front and a side face (an "L" shape), or only a side face is visible, the dominant cause of failure for a naive single-face normal estimate at large articulation. The four 90° heading candidates implied by $\theta^\star$ are disambiguated by choosing the one closest to the previous filtered estimate and most consistent with the dimension prior, yielding a trailer longitudinal direction $\hat{\mathbf{d}}_t = (\cos\theta^\star, \sin\theta^\star)$.

Articulation angle and filtering. The raw articulation angle is the signed angle between the truck-forward axis $\hat{\mathbf{x}} = (1, 0)$ and $\hat{\mathbf{d}}_t$,

$$\tilde{\gamma} = \operatorname{atan2}\big(\hat{x}_1 \hat{d}_{t,2} - \hat{x}_2 \hat{d}_{t,1},\ \hat{\mathbf{x}} \cdot \hat{\mathbf{d}}_t\big), \tag{6.4}$$

to which a field-calibration sign s and offset γ_0 are applied and the result clamped to the operational range, $\gamma = \operatorname{clip}(s\,\tilde{\gamma} + \gamma_0,\, -\gamma_{\max},\, \gamma_{\max})$. The sign convention is chosen so that the published value matches Autoware's $\gamma = \psi_{\text{truck}} - \psi_{\text{trailer}}$. The raw per-scan estimate is then passed through a constant-angular-velocity Kalman filter [88] on the state $[\gamma,\ \dot{\gamma}]$, with Mahalanobis innovation gating to reject outlier scans, and a short moving-average is maintained in parallel as a diagnostic. The filtered angle and its rate populate the `hitch_angle` and `hitch_rate` fields of the published `TrailerState`.

This onboard, geometry-plus-filter estimator is the actual mechanism behind the `/vehicle/trailer_state` signal. It is not an external measurement. Its noise characteristics, and the way they are handled, are revisited in the gap analysis (Section 6.4.3).

6.3.5 Trailer State Estimation

Because the trailer carries no independent localization, its full pose is reconstructed from the tractor's localized kinematic state and the estimated articulation angle. Given the tractor's rear-axle position and heading $(\mathbf{p}_{\text{truck}},\, \psi_{\text{truck}})$ from `/localization/`

`kinematic_state`, the hitch-point position is

$$\mathbf{p}_{\text{hitch}} = \mathbf{p}_{\text{truck}} + d_h \begin{bmatrix} \cos\psi_{\text{truck}} \\ \sin\psi_{\text{truck}} \end{bmatrix}, \tag{6.5}$$

where d_h is the (small, near-zero) signed hitch offset relative to the tractor rear axle. The trailer heading follows directly from the articulation angle, $\psi_{\text{trailer}} = \psi_{\text{truck}} - \gamma$, and the trailer rear axle is located at

$$\mathbf{p}_{\text{trailer}} = \mathbf{p}_{\text{hitch}} - L_t \begin{bmatrix} \cos\psi_{\text{trailer}} \\ \sin\psi_{\text{trailer}} \end{bmatrix}, \tag{6.6}$$

where $L_t \approx 2.8\,\text{m}$ is the trailer wheelbase. This reconstruction mirrors the hitch-kinematics model derived in Chapter 3, confirming that the simulation formulation is directly reusable in the deployment stack, and it provides the trailer-referenced tracking errors that form part of the policy observation.

6.3.6 Actuator-Delay Compensation: Smith Predictor

The simulation trains the policies against an idealized actuator that responds instantaneously to commands, whereas the physical steering and drive systems exhibit a first-order lag of the order of 50–150 ms. A policy trained for instantaneous response will, when confronted with a delayed actuator, repeatedly over-correct and excite a lag-induced oscillation. To reconcile the two, a Smith-predictor-style compensator is implemented that lets the delay-free-trained policy act on a delay-compensated state estimate.

The compensator maintains two "shadow" instances of the same kinematic `StateSpaceTractorTrailer` model used in simulation: a delay-free shadow with zero actuation time constant, and a delayed shadow whose time constant matches the measured hardware lag. On each control tick both shadows are advanced with the command currently propagating through the actuator pipeline. The discrepancy between the measured vehicle state and the delayed shadow forms a residual,

$$\mathbf{r}_k = \mathbf{x}_k^{\text{meas}} - \mathbf{x}_k^{\text{delayed}}, \tag{6.7}$$

which captures model mismatch beyond pure delay (integration scheme, ground-contact effects, friction). The state presented to the policy is the delay-free shadow corrected by this residual,

$$\hat{\mathbf{x}}_k = \mathbf{x}_k^{\text{free}} + \mathbf{r}_k, \tag{6.8}$$

so that the policy "sees" an effectively delay-free vehicle while the residual keeps the prediction anchored to reality. The shadows are re-synchronized to the measured state on reset and whenever a large state jump is detected. The compensator is implemented and enabled in the deployment launch configuration, though its shadow time constants still require per-platform tuning before the loop can be closed on hardware. Combined with the explicit steering-rate limit it forms the primary mitigation for the actuation gap discussed in Section 6.4.2.

Learned dynamics model (in progress). The Smith predictor's fidelity is bounded by the accuracy of the analytical kinematic shadow. Work is in progress to replace, or augment, that shadow with a neural-network approximation of the true system dynamics. A data pipeline converts recorded driving logs (rosbags) into time-aligned 10 Hz state–action–next-state datasets (deriving the trailer pose from the truck pose and the estimated hitch angle where it is not directly recorded) and a multilayer-perceptron dynamics model (`DynamicsMLP`) is trained on these datasets. The learned model is evaluated both by one-step prediction error against held-out logs and by closed-loop rollout divergence from the kinematic model, the latter serving to quantify and localize the sim-to-real dynamics gap. This component is not yet part of the live control loop and is reported here as ongoing work rather than a validated result.

6.4 Sim-to-Real Gap Analysis

Transferring a simulation-trained policy to physical hardware introduces discrepancies between the simulated and real-world dynamics. The following gap sources were identified and addressed during the deployment process.

6.4.1 Vehicle Scale and Parameterization

The most fundamental difference between the simulation studies and the physical platform is one of scale. The simulation of chapter 3 is parameterized for a full-size shunt truck (table 3.1), whereas the deployment vehicle is the compact Agilex Hunter SE (wheelbase 0.65 m, maximum steering angle 0.436 rad), with under a quarter of the wheelbase and half the steering range of the reference terminal tractor. Rather than transferring the shunt-truck-scale policy onto the smaller vehicle and relying on it to generalize across scale, the deployment closes this gap by construction. The model structure of chapter 3 is retained unchanged and re-parameterized to the Hunter's wheelbase, steering limit, and speed range, and a fresh policy is trained against that re-parameterized model. Because the governing equations are identical and only the parameter set changes, the observation and action interfaces the policy was trained against are preserved, and the scale mismatch is removed before deployment rather than being left as a residual gap to be compensated at run time.

6.4.2 Actuation Dynamics

The simulation model assumes instantaneous steering response, whereas the physical vehicle exhibits a first-order steering lag introduced by the CAN command latency and the Hunter SE's internal steering servo. This is the highest-impact discrepancy for a policy trained on instantaneous actuation. It is mitigated by the Smith-predictor compensator of Section 6.3.6, which presents a delay-compensated state to the policy, and by a steering-rate limit applied to the commanded output, consistent with the $\dot{\delta}_{\max}$ constraint used during simulation training.

6.4.3 Hitch-Angle Sensing

In simulation, the articulation angle γ is available exactly from the vehicle state. On hardware it is estimated from LiDAR returns and is therefore subject to noise from sparse or partial returns on the trailer face, mis-segmentation of the region of interest, and momentary loss of the face at large articulation. These error modes are addressed within the estimator itself rather than by a downstream filter. The rectangle-fitting formulation (Equation 6.3) is robust to partial face visibility, and the constant-angular-velocity Kalman filter with innovation gating rejects outlier scans and smooths the published signal. If the trailer face is lost entirely the estimator withholds new estimates, and the bridge continues from the last published value until the face is reacquired. No automatic disengagement on stale data is currently implemented.

6.4.4 Path Distribution

The policies are trained on smooth, spline-generated reference paths, whereas the deployment reference is the centreline of a Lanelet2 lane, which is piecewise-linear and may contain comparatively sharp vertices. Re-sampling the centreline does not fully reconcile the difference in path-curvature distribution between training and deployment, and this mismatch is a contributing factor to any residual tracking error observed on hardware. It is noted here as a characterization of the gap rather than something fully eliminated in the current deployment.

6.4.5 Jackknife Prevention

The simulation introduced a terminal episode condition when $|\gamma| > 90°$ but provided no explicit recovery mechanism, since the episode simply resets. Rather than adding a separate runtime supervisor on the vehicle, jackknife resilience is instilled during training, through two complementary measures. First, a recovery-scenario reset distribution spawns a fraction of reverse episodes already in the near-jackknife region (articulation magnitudes of roughly 0.35–0.80 rad, i.e. 20–45°), so the policy is repeatedly exposed to high-articulation states its nominal training distribution rarely visits and must learn to steer the trailer back toward alignment. Second, an anti-jackknife override is applied within the training environment. Whenever the articulation exceeds approximately 0.7 rad (40°), the policy's steering command is replaced by a maximum counter-steer that drives the articulation back toward zero. This prevents the policy from learning that oscillatory, near-jackknife behaviour is survivable, since every dangerous excursion is corrected during training and chattering no longer pays off. No dedicated runtime recovery override is currently part of the deployed bridge. On hardware, jackknife avoidance therefore relies on the behaviour learned through these measures together with the operational $|\gamma| \leq 80^\circ$ limit.

6.4.6 Localization Dependency

The simulation provides ground-truth vehicle pose at every step. The hardware stack instead depends on the Autoware localization pipeline, which fuses RTK-GNSS, gyro-based odometry, and LiDAR NDT scan matching through an extended Kalman filter. GNSS outages or localization divergence therefore represent a failure mode that does not exist in simulation. The current deployment treats localization failure as an emergency-stop condition.

6.5 High-Fidelity Geographic Simulation

The 2D Pygame environment of chapter 3 is deliberately abstract. It captures the tractor-trailer kinematics and lane geometry but not the three-dimensional and photometric structure that the deployed perception and localization stack actually consumes. A complementary, higher-fidelity simulation environment was therefore built to exercise the full Autoware stack (LiDAR-based NDT localization, perception, and the RL bridge) inside a geographically faithful "digital twin" of a real operating site as a complement to the physical trials.

This environment is built on the Gazebo simulator [89], extending the open-source Rudis Laboratories georeferenced world-generation toolchain [90]. Real locations are reconstructed as textured 3D meshes authored in Blender [91], whose physically based materials are baked into texture maps for efficient real-time rendering. Building on that toolchain, the target distribution-centre site was rebuilt in Blender directly from public geographic data, with both layers imported through the Blender GIS plugin. Building footprints were pulled from OpenStreetMap [92] and extruded to form the 3D building meshes, and these were draped over aerial orthoimagery from the ESRI World Imagery basemap, replacing the lower-resolution ground textures of the original toolchain with the higher-resolution ESRI imagery (fig. 6.1). Because the reconstruction is driven entirely by the OpenStreetMap footprints and the ESRI imagery layer, the same procedure generates a Gazebo world for any real distribution centre from its coordinates alone, rather than being tied to a single hand-authored site. Each world is anchored to GIS coordinates through the SDF `spherical_coordinates` element, which fixes the simulation origin to a real latitude, longitude, and elevation. The templated world generator additionally computes the sun's azimuth and elevation from the site coordinates and a chosen UTC date and time, so that lighting and shadows match the real location and time of day. The `electrans` tractor and trailer models are then spawned into these worlds, so that the same articulated vehicle can be driven through a virtual replica of the target environment (figs. 6.2 and 6.3).

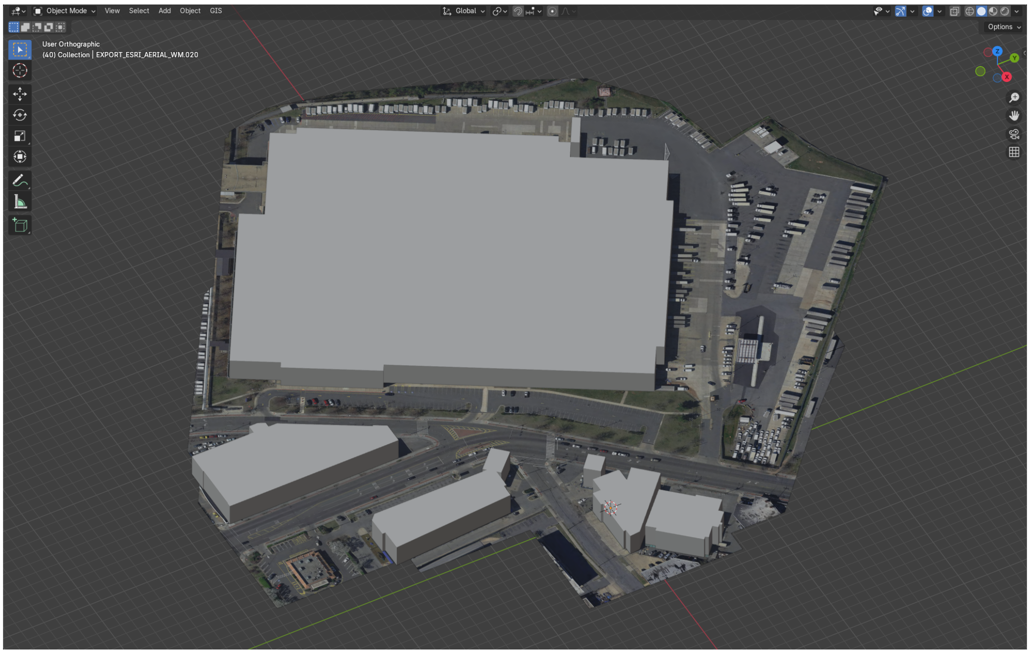


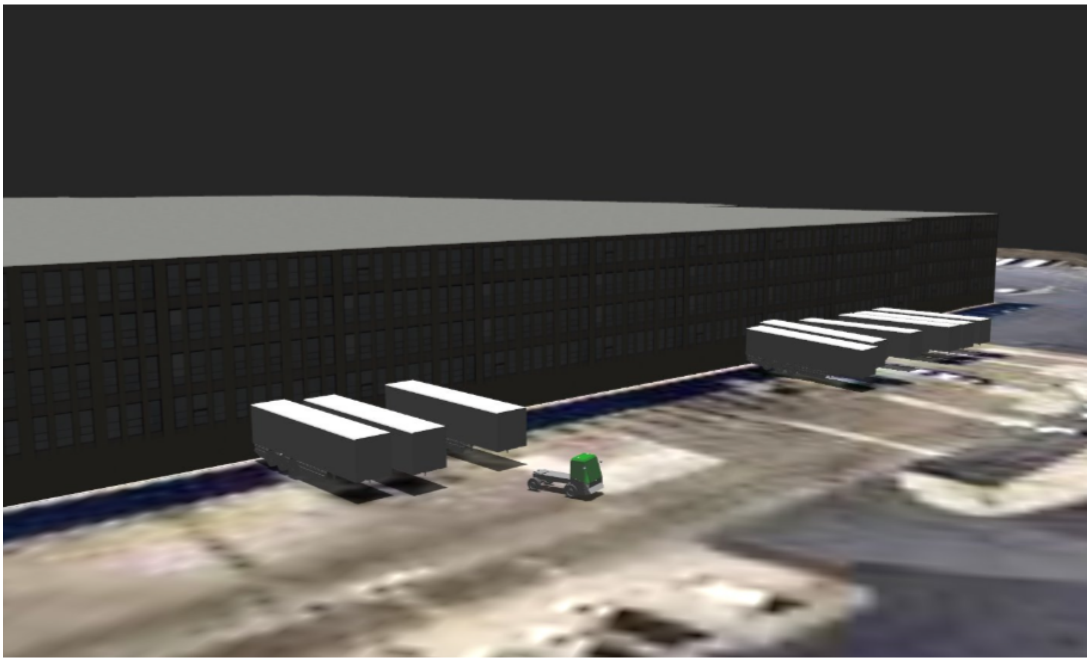
Figure 6.1: Reconstruction of the target distribution-centre site in Blender. OpenStreetMap building footprints are extruded to form the 3D building meshes and draped over high-resolution aerial orthoimagery from the ESRI World Imagery basemap, both layers imported through the Blender GIS plugin. The same GIS-driven procedure reconstructs any real distribution centre from its geographic coordinates.

Figure unavailable

Reconstructed distribution centre exported to a georeferenced Gazebo world.

`figures/gazebo/gazebo_dc_world.png`

(a) The reconstructed site exported to a georeferenced Gazebo world.



(b) The `electrans` tractor-trailer spawned at a loading dock within the world.

Figure 6.2: The georeferenced site running in Gazebo. The exported world (a) preserves the real building geometry and aerial ground texture, and the `electrans` articulated vehicle (b) is spawned into it so that the same policies evaluated in simulation can be exercised against the full Autoware sensing and localization stack.

Because the world is georeferenced and rendered in three dimensions, this environment directly targets several of the sim-to-real gaps catalogued above. A LiDAR sensor model produces point clouds against real building and ground geometry, so NDT scan matching and the onboard hitch-angle estimator (section 6.3.4) can be exercised against realistic returns; the geographic anchoring lets the GNSS and map frames be reproduced consistently with the physical site; and the photorealistic rendering makes it possible to evaluate camera-based perception that the abstract 2D environment cannot represent. It is also the natural setting in which to prototype the infrastructure sensor nodes of section 6.6, since virtual LiDAR units can be placed at arbitrary surveyed locations in the georeferenced world. The environment and its GIS-driven world-generation pipeline are complete. What remains is the quantitative characterization of policy behaviour within it, which is identified as future work in chapter 7.

6.5.1 Articulated Vehicle Model

The articulated vehicle spawned into the georeferenced worlds is the `electrans` tractor-trailer, shown in fig. 6.3. It is defined in the conventional ROS/Gazebo manner as a URDF assembled from `xacro` macros, rather than as a hand-written SDF model. The tractor shares its entire `xacro` description with the Agilex Hunter model used elsewhere in the project. The same parameterized macros define an Ackermann steering geometry (a pair of revolute front-steering links bounded by the maximum steering angle), continuous-rotation rear wheels coupled by a mimic joint, the link inertials, and the Gazebo friction and `ros2_control` blocks. The two vehicles differ only in scale and in the body mesh. The Hunter's compact chassis mesh is replaced by a cab-over terminal-tractor body, and the model is scaled accordingly. Reusing a single `xacro` description in this way keeps the simulated tractor kinematically consistent with the physical Hunter platform on which the policies are deployed, while presenting the larger visual form of the terminal tractor this thesis targets.

The trailer is coupled to the tractor through a revolute hitch joint at the tractor rear axle, matching the hitch convention of the kinematic model in chapter 3. It is modelled as a single box body carried on one axle and reuses the same wheel meshes and inertial macros as the tractor, so that the complete articulated system is expressed through the shared macro library. Because the model is authored as standard xacro, it drops directly into both the Gazebo digital twin of this section and the wider Autoware simulation tooling without modification.

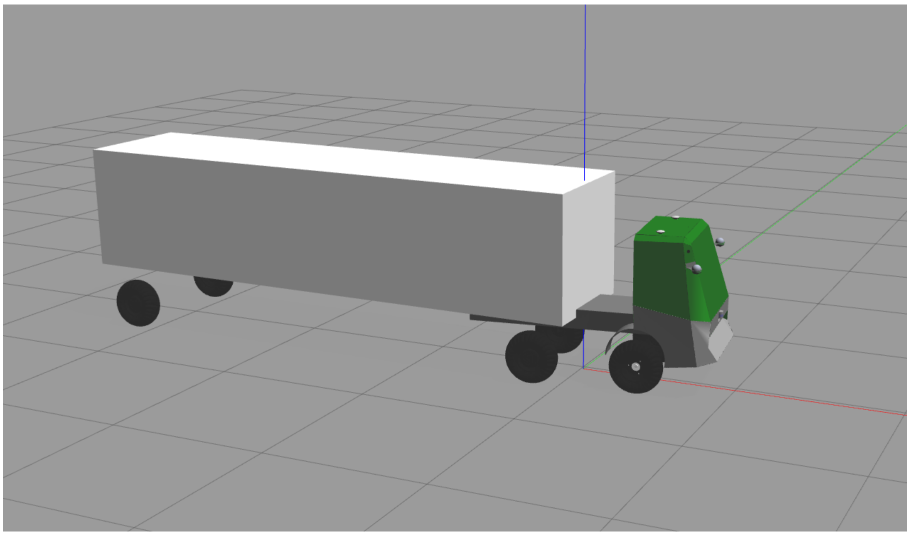


Figure 6.3: The `electrans` tractor-trailer model. The tractor reuses the Agilex Hunter `xacro` description (Ackermann front steering, continuous rear wheels, and `ros2_control` blocks), rescaled and given a cab-over terminal-tractor body mesh. The box trailer is attached through a revolute hitch joint at the tractor rear axle.

6.6 Infrastructure Sensor Nodes

A structural limitation of the onboard sensing arrangement is that the trailer occludes the region directly behind it. The tractor-mounted LiDAR cannot see past the trailer, and instrumenting the trailer itself is undesirable. It would add cabling, power, calibration, and a wireless link to an otherwise passive, interchangeable body, and would have to be repeated for every trailer. This is most acute during reverse manoeuvring, precisely when visibility of the space behind the trailer matters most.

This limitation is addressed with off-vehicle infrastructure sensor nodes, fixed LiDAR units placed at surveyed vantage points around the operating area, each providing a bird's-eye view of obstacles (including the region behind the trailer) without any sensor being mounted on the trailer. Each node is calibrated into a shared global frame so that its detections can be expressed in the same coordinates as the onboard perception output and fused into Autoware's perception and occupancy-grid representation, yielding a persistent, occlusion-resilient view of the obstacles around the articulated vehicle that complements the limited onboard field of view.

This capability is implemented and running in the laboratory as the indoor-perception stack. Each infrastructure node runs Euclidean clustering on its own LiDAR returns to detect candidate objects. An offboard fusion pipeline transforms every node's detections into the shared global frame, merges detections that refer to the same physical object across nodes by their overlapping footprint polygons, and time-aligns the streams against each node's modelled latency before passing the merged detections to an extended-Kalman-filter multi-object tracker. The tracker emits a single stable, deduplicated obstacle list for the whole monitored area.

Because the laboratory sensor nodes and their perception stack run under ROS1 (Noetic) while the vehicle runs under ROS2 (Humble), the two are joined by a dedicated interface node. It subscribes to the ROS1 perception outputs and republishes the fused obstacle list over ROS2, where it is consumed by the vehicle's Autoware perception and occupancy-grid representation and thus reaches the RL bridge. Critically, this interface node runs offboard on the laboratory compute rather than on the vehicle. The vehicle's onboard Jetson therefore runs only ROS2, and never has to host a dual ROS1/ROS2 stack. The ROS1 dependency is confined entirely to the off-vehicle infrastructure, which both simplifies the embedded software image and keeps the version-sensitive ROS1 perception packages off the resource-constrained onboard computer.

In laboratory operation the infrastructure nodes recover the region directly behind the trailer that the onboard LiDAR cannot observe, extending obstacle coverage to [TODO]% of the vehicle's immediate surroundings during reverse manoeuvres, at an added end-to-end latency of [TODO] ms relative to onboard-only perception.

Figure unavailable

Diagram of the infrastructure LiDAR sensor nodes providing a bird's-eye view of obstacles behind the trailer.

`figures/system_architecture/infrastructure_sensor_nodes`

Figure 6.4: Infrastructure sensor-node architecture: fixed off-vehicle LiDAR units give a bird's-eye view of obstacles around the articulated vehicle, covering the region behind the trailer that the onboard sensors cannot observe. Their detections are fused and tracked offboard and delivered to the vehicle over ROS2, so the onboard computer runs only ROS2.

6.7 Deployment Results

Hardware validation trials were conducted on [TODO: describe test location and scenario]. The forward policy achieved [TODO: describe forward lane-following behaviour], with a mean trailer cross-track error of [TODO] m and articulation maintained within $|\gamma| < [TODO]^\circ$ throughout nominal operation. Reverse manoeuvring [TODO: describe outcome]. [TODO: report the effect of enabling the Smith predictor on oscillation and tracking once tuned.]

6.8 Summary

This chapter demonstrated that the tractor-trailer control architecture developed in simulation can be ported to a physical platform by executing the trained RL policies directly through a bridging layer, rather than by re-deriving an analytical controller on the vehicle. The deployment retains Autoware's standard localization and perception while replacing planning and control with an RL bridge that runs the forward and reverse policies at 10 Hz over the same observation and action interfaces used in training. The principal engineering contributions are this RL bridge and its integration into a stripped-down Autoware stack, the onboard LiDAR hitch-angle estimator that recovers the articulation angle by rectangle fitting and Kalman filtering without any trailer-mounted sensor, and the Smith-predictor compensator (with an in-progress learned dynamics model) that reconciles delay-free-trained policies with a lagging physical actuator. The gap analysis confirms that the simulation captures the dominant dynamics relevant to controller design, while identifying actuation delay, hitch-angle sensing, path-distribution mismatch, and localization reliability as the primary points of divergence between the simulated and physical systems. Finally, the infrastructure sensor nodes, implemented and running in the laboratory, provide a route to occlusion-resilient perception behind the trailer without adding sensors to the trailer itself.

Chapter 7

Conclusion

7.1 Summary of Findings

This thesis presented a reinforcement learning framework for articulated autonomous vehicle control. The key contributions demonstrated are:

- a combined dynamic tractor and kinematic trailer model implemented in a custom Gymnasium simulation environment;
- a vision-based observation space in which compact BEV crops derived from the Pygame simulation are distilled into the vehicle's path geometry by a frozen encoder, enabling spatial perception of lane boundaries and obstacles;
- a TD3 reinforcement learning agent trained on lane-following and obstacle avoidance tasks with reward formulations that penalize cross-track error, heading error, hitch articulation angle, terminal failures, and close proximity to lane boundaries or obstacles;
- comparison of TD3 performance against tuned pure-pursuit, PID, and MPC classical control baselines on matched articulated path-tracking tasks;
- a sim-to-real deployment (Chapter 6) that runs the trained forward and reverse policies directly on a physical Agilex Hunter SE through a stripped-down Autoware stack, supported by onboard LiDAR-based hitch-angle estimation and a Smith predictor for actuator-delay compensation.

The integration of visual perception via BEV imagery broadens the control problem beyond compact vector-state tracking. Rendered BEV images give the agent spatial context about lane geometry and obstacle positions that vector-state observations alone cannot provide. An observation and training-method ablation found that inserting a raw image branch into the policy, whether learned end to end or pre-trained as an autoencoder or a UNet, does not by itself produce a trainable vision policy. The approach that does is to distill the BEV image into the vehicle's path geometry with a frozen encoder and train the policy on that geometry, which recovers the learning behaviour of the engineered-state baseline.

From a research-impact perspective, the thesis is strongest when interpreted not only as an RL training exercise but as a reproducible benchmark and systems study for articulated autonomy. The combination of matched observation ablations, tuned classical baselines, fixed-scenario benchmarking, and explicit articulation-stability metrics provides a framework for asking which sensing and control choices are actually useful for tractor-trailer path tracking. This framing is more defensible and more reusable than reporting isolated policy demonstrations.

7.2 Limitations

The study is subject to the following constraints:

- The bulk of the experimental evaluation is conducted within the custom 2D Pygame/Gymnasium simulation environment. A hardware deployment exists (Chapter 6), but quantitative sim-to-real performance has not yet been characterized, and behaviour in higher-fidelity physics simulators remains untested.
- The 2D simulation approximates the vehicle dynamics but does not capture all effects present in a real tractor-trailer system, such as load transfer, tire saturation at high slip angles, or hitch joint compliance.
- The current framework assumes a fixed payload. Hitch dynamics change significantly with varying trailer mass.
- Obstacle avoidance is limited to static obstacles. Dynamic obstacle handling is not addressed.
- Generalization beyond the training distribution has not yet been fully quantified. Robustness to sensor degradation and stronger path-distribution shift remains to be measured explicitly. Actuator delay is actively mitigated on hardware by the Smith predictor of Chapter 6, but its effect on closed-loop tracking has not yet been measured.
- The hardware deployment of Chapter 6 runs the trained policies on the Agilex Hunter SE through the ROS2/Autoware integration, but a full characterization of sim-to-real performance across the complete range of training tasks remains as future work.

7.3 Future Work

Several directions remain open for future investigation:

- **Higher-fidelity simulation**: Validating the trained policies in a physics-based 3D simulator with a detailed model of the tractor-trailer system, including realistic tire and hitch dynamics. This is already underway as the georeferenced Gazebo "digital-twin" environment of section 6.5, which reconstructs real sites from Blender-authored, GIS-anchored worlds and exercises the full Autoware perception and localization stack. Completing it and quantifying the policies' behaviour within it is the natural next step.

- **Hardening the hardware deployment**: The trained TD3 policies are already deployed directly on the Hunter SE through the RL bridge of Chapter 6, with actuator-delay compensation provided by the Smith predictor. The next steps are to collect quantitative sim-to-real results across the full task set, complete the in-progress learned neural-network system-dynamics model that augments the Smith predictor, introduce domain randomization during training to narrow the remaining sim-to-real gap, and add a supervised safety-fallback layer.

- **Infrastructure sensing**: The off-vehicle infrastructure LiDAR sensor nodes proposed in Chapter 6 are implemented and running in the laboratory, providing an occlusion-resilient bird's-eye view of obstacles behind the trailer without instrumenting the trailer itself and fusing their detections into the onboard perception stack. The remaining work is to characterize their coverage and latency quantitatively and to extend them from the laboratory to the deployment site.

- **Planner-controller baselines for obstacle tasks**: Comparing obstacle-avoidance RL policies against complete classical navigation stacks, so that end-to-end navigation is evaluated on a fair task definition rather than against controller-only baselines.

- **Robustness benchmarks**: Adding out-of-distribution test sets with stronger curvature, perturbed initial poses, observation noise, and parameter mismatch, so that the thesis can quantify generalization rather than only nominal performance.

- **Failure-mode analysis**: Building a structured taxonomy of controller failures, including oscillation growth, corner-cutting, hitch divergence, and obstacle-clearance collapse, to support safety-oriented comparison rather than relying on averages alone.

- **Control-aligned visual pretraining**: Pretraining BEV encoders to predict control-relevant quantities such as cross-track error, heading error, hitch angle, curvature, or expected reward, rather than relying only on pixel reconstruction.

- **Sequence-model policies for tractor-trailer control**: Applying transformer-based offline control methods to the articulated vehicle setting, with training data generated from the TD3 expert policy and augmented with high-error recovery trajectories.
- **Multi-trailer systems**: Extending the kinematic model and RL policy to double or B-train configurations.
- **Online adaptation**: Updating the policy in real time to compensate for varying payloads and road conditions.
- **Control Barrier Function overlays**: Implementing CBFs as a hard safety layer that overrides the RL agent when a collision or jackknife is imminent.
- **Dynamic obstacles**: Extending the obstacle avoidance environment to include moving obstacles and integrating predictive collision avoidance.

Appendix A

Classical Controller Implementation

The deep reinforcement-learning policy is benchmarked against four classical path-tracking controllers. This appendix documents their implementation: the exact control laws, the forward and reverse variants, the way each controller was tuned, and the harness used to evaluate them. The four controllers are chosen to span the classical control spectrum, from a purely geometric law through to constrained optimal control with preview:

- **Pure Pursuit**, a geometric proportional law;
- **PID**, classical error feedback with an integral term;
- **LQR**, optimal static linear feedback on the error dynamics;
- **Kinematic LTV MPC**, optimal constrained control with preview.

All four are implemented in the batched simulator (`tractor_trailer_rl_cupy`) so that they run inside the same vectorized environment as the learned policies. Crucially, every controller emits the same action as the RL agent, a bounded steering rate, so the comparison between classical and learned control is made on identical terms.

A.1 Common framework

A.1.1 Observation layout and shared action mapping

Each controller is a function of the observation vector already produced by the environment. For the trailer tasks this vector is

$$o = [\, s,\ \gamma,\ e_y,\ e_\psi,\ e_{y,t},\ e_{\psi,t},\ \kappa,\ \ldots], \tag{A.1}$$

where s is the current steering angle, γ the hitch (articulation) angle, and the error pairs (e_y, e_ψ) and $(e_{y,t}, e_{\psi,t})$ are the cross-track and heading errors of the tractor and of the trailer respectively; κ is the reference path curvature at the lookahead point. The trailing entries carry additional task-specific channels (for example lidar returns) that the classical controllers do not use.

Every controller computes a desired steering angle δ_{des} and then converts it to the environment's steering-rate action through a common saturating map,

$$\dot{\delta} = \text{clip}\left(\frac{\delta_{\text{des}} - s}{\Delta t},\ -\dot{\delta}_{\text{max}},\ +\dot{\delta}_{\text{max}}\right), \tag{A.2}$$

where Δt is the simulation step and $\dot{\delta}_{\text{max}}$ the steering-rate limit. Because the learned policy also outputs $\dot{\delta}$, and eq. (A.2) applies to all four baselines identically, the classical and learned controllers act through exactly the same interface.

A.1.2 Forward and reverse tracking

Each controller has a forward and a reverse variant, reflecting a structural asymmetry of the articulated vehicle. Driving forward, the tractor leads and the trailer follows passively, so it suffices to regulate the tractor errors (e_y, e_ψ) towards zero. Driving in reverse, the trailer leads and the coupling inverts. The hitch angle γ becomes open-loop unstable (small articulation errors grow and the vehicle jackknifes), so the reverse controllers regulate the trailer errors $(e_{y,t}, e_{\psi,t})$ and must additionally include an explicit hitch-stabilizing action. This asymmetry is why the reverse variants below are not simply the forward laws with a sign change, and why their gains had to be tuned separately (appendix A.6).

A.1.3 Vehicle parameters

The benchmark uses the full-size shunt-truck configuration. The parameters that enter the controllers are summarized in table A.1. Here L_1 is the tractor wheelbase, L_2 the effective trailer length (hitch to trailer axle), and V the fixed longitudinal speed held constant throughout the fixed-speed path-tracking experiments. The large length ratio $L_2/L_1 \approx 4.2$ is what makes the reverse hitch dynamics difficult to stabilize.

Table A.1: Shunt-truck parameters used by the classical controllers.

Symbol	Quantity	Value
L_1	Tractor wheelbase	2.95 m
L_2	Trailer length (hitch to axle)	12.5 m
V	Fixed longitudinal speed	5.0 m/s
Δt	Simulation step	0.1 s
$\delta_{\max}$	Steering-angle limit	0.87 rad ($\approx 50°$)
$\dot{\delta}_{\max}$	Steering-rate limit	25°/s

In the reverse variants the speed enters with a sign, $V < 0$. Because the environment's tractor yaw response to steering does not itself flip sign in reverse, the along-path kinematics use the magnitude $|V|$ while the yaw dynamics retain the signed V. This distinction is carried consistently through the LQR and MPC models below.

A.2 Pure Pursuit

The pure-pursuit controller [18] is a fixed proportional geometric law. Forward, it combines a curvature feed-forward with proportional cross-track and heading feedback on the tractor errors,

$$\delta_{\text{des}} = -\big(k_{\text{ff}} \arctan(L_1 \kappa) + k_y\, e_y + k_\theta\, e_\psi\big). \tag{A.3}$$

Reverse, it regulates the trailer errors and adds an explicit hitch-stabilizing term $k_{\text{hitch}}\, \gamma$ to counter the open-loop-unstable articulation,

$$\delta_{\text{des}} = k_{\text{hitch}}\, \gamma + k_{\text{ff}}\, \kappa + k_y\, e_{y,t} + k_\theta\, e_{\psi,t}. \tag{A.4}$$

Both laws are stateless functions of the observation, so the controller is fully vectorized across the parallel environments. Besides serving as a benchmark, the tuned pure-pursuit controller doubles as the reference ("teacher") signal for the guided reward and as a feasibility oracle. A path that a well-tuned pure-pursuit controller can complete is deemed geometrically feasible.

A.3 PID

The PID controller extends the proportional law with an integral term that removes steady-state cross-track offset. The integral is accumulated per environment and clamped to prevent windup, and it is reset at episode boundaries. Forward,

$$\delta_{\text{des}} = -k_{\text{ff}}\,\kappa - K_p\,e_y - K_i\int e_y - K_d\,e_\psi + K_{p,t}\,e_{y,t}, \tag{A.5}$$

and reverse (regulating the trailer errors, with the hitch stabiliser),

$$\delta_{\text{des}} = k_{\text{hitch}}\,\gamma + k_{\text{ff}}\,\kappa + K_p\,e_{y,t} + K_i\int e_{y,t} + K_d\,e_{\psi,t}. \tag{A.6}$$

The derivative gain K_d multiplies the heading error, so it acts as a heading-damping term rather than a numerical time derivative. The integrator uses the anti-windup update

$$I \leftarrow \operatorname{clip}\big(I + e\,\Delta t,\ -I_{\max},\ +I_{\max}\big), \tag{A.7}$$

where e is the regulated cross-track error (e_y forward, $e_{y,t}$ reverse). Because the controller carries this per-environment state, the evaluation loop passes a termination mask so that I is zeroed for any environment that resets on the current step.

A.4 LQR

The LQR baseline is the canonical optimal-linear controller, static feedback $u = -Kx$ on a linear model of the tracking-error dynamics, plus a curvature feed-forward [67]. The error state is

$$x = [\,e_{\text{lat}},\ e_{\text{head}},\ \gamma\,]^\top, \tag{A.8}$$

the lateral and heading error of the leading unit (tractor forward, trailer reverse) together with the hitch angle. The input is the steering angle $u = \delta$. The continuous error dynamics $\dot{x} = A_c x + B_c u$ are direction-specific. Forward,

$$A_c = \begin{bmatrix} 0 & V & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -\dfrac{V}{L_2} \end{bmatrix}, \qquad B_c = \begin{bmatrix} 0 \\ \dfrac{V}{L_1} \\ \dfrac{V}{L_1} \end{bmatrix}, \tag{A.9}$$

and reverse, where the leading trailer progresses along the path at speed $|V|$ while the yaw dynamics keep the signed V,

$$A_c = \begin{bmatrix} 0 & |V| & 0 \\ 0 & 0 & \dfrac{V}{L_2} \\ 0 & 0 & -\dfrac{V}{L_2} \end{bmatrix}, \qquad B_c = \begin{bmatrix} 0 \\ 0 \\ \dfrac{|V|}{L_1} \end{bmatrix}. \tag{A.10}$$

The pair (A_c, B_c) is discretized by zero-order hold at the step Δt to give (A_d, B_d). With state and input weights $Q = \mathrm{diag}(q_{\mathrm{lat}}, q_{\mathrm{head}}, q_\gamma)$ and R, the infinite-horizon gain follows from the discrete algebraic Riccati equation. Its stabilizing solution P yields

$$K = \left(R + B_d^\top P B_d\right)^{-1} B_d^\top P A_d. \tag{A.11}$$

Because the speed is constant, K is computed once at construction and then applied to every environment, so the controller stays fully vectorized. The command adds a steady-state feed-forward that holds the vehicle on a constant-curvature arc: a feed-forward steer $\delta_{\mathrm{ff}} = L_1 \kappa$ and a hitch reference $\gamma_{\mathrm{ref}} = \mathrm{sign}(V)\, L_2\, \kappa$, giving

$$\delta = \delta_{\mathrm{ff}} - K\,(x - x_{\mathrm{ref}}), \qquad x_{\mathrm{ref}} = [\,0,\ 0,\ \gamma_{\mathrm{ref}}\,]^\top, \tag{A.12}$$

which is then saturated to $\pm\delta_{\max}$ and mapped to a steering rate through eq. (A.2). The reverse weights place a heavy penalty on control effort and hitch deviation, which is what damps the otherwise unstable reverse plant.

A.5 Kinematic LTV MPC

The final baseline is a linear time-varying model-predictive controller [6] that adds a finite preview horizon and explicit input constraints. The MPC uses the same error state $x = [\,e_{\mathrm{lat}}, e_{\mathrm{head}}, \gamma\,]^\top$ as the LQR but plans over an N-step horizon.

A.5.1 Prediction models

Forward, the tractor leads, so the MPC tracks the tractor errors with the steering angle δ as the input, giving the same (A_c, B_c) as the forward LQR model (A.9). Reverse, servoing the trailer pose with the unstable hitch dynamics inside the optimization causes the long shunt trailer to oscillate and jackknife. The reverse controller therefore uses a flatness / virtual-input reformulation (following the Autoware trailer LTV-MPC): the QP plans the trailer as a stable bicycle driven by a virtual trailer steer δ_T, and the hitch enters only through an inner loop that has been closed to the stable first-order dynamics $\dot{\gamma} = K(\delta_T - \gamma)$,

$$A_c = \begin{bmatrix} 0 & |V| & 0 \\ 0 & 0 & \dfrac{V}{L_2} \\ 0 & 0 & -K \end{bmatrix}, \qquad B_c = \begin{bmatrix} 0 \\ 0 \\ K \end{bmatrix}. \tag{A.13}$$

The reference curvature enters both models as a measured disturbance through $E_c = [\,0,\ -|V|,\ 0\,]^\top$. After the QP returns the virtual input, an analytic inner map realises it on the physical tractor and supplies the anti-jackknife counter-steer,

$$\delta_f = \arctan\left(\frac{L_1}{L_2}\,\mathrm{sign}(V)\,\sin\gamma + \frac{L_1 K}{|V|}\,(\delta_T - \gamma)\right). \tag{A.14}$$

The $1/|V|$ factor is what flips the feedback sign in reverse and produces the stabilizing counter-steer.

A.5.2 Condensed quadratic program

The continuous models are discretized by zero-order hold and the state trajectory is condensed onto the input sequence, so that the optimization variable is the vector of N future steering inputs. With the stacked weight $\bar{Q} = I_N \otimes Q$, a control penalty R, an input-rate penalty R_d, and the first-difference operator $T = I - I^{(-1)}$ (so that T forms Δu), the cost Hessian is

$$H = 2\big(B_u^\top \bar{Q}\, B_u + R\, I + R_d\, T^\top T\big), \tag{A.15}$$

where B_u is the condensed input-to-state map. The problem is subject to a steering-rate box on the increments, $|\Delta u| \leq \dot{\delta}_{\max}\,\Delta t$, and a steering box $|u| \leq \delta_{\max}$. It is solved per environment with OSQP. The rate-penalty term $R_d\, T^\top T$ is weighted heavily in reverse so the controller does not chase cross-track measurement ripple. The overall per-step procedure is summarized in algorithm 1.

Algorithm 1 Kinematic LTV MPC step (per environment)

1: **Precompute (once):** discretize (A_c, B_c, E_c) by ZOH; build the condensed maps A_x, B_u, E_w and the constant Hessian H from (A.15).
2: **On episode reset:** clear the warm-start input $u_{\text{prev}} \leftarrow 0$.
3: Read $x_0 = [\,e_{\text{lat}}, e_{\text{head}}, \gamma\,]$ and curvature κ from the observation.
4: Form the linear cost term from x_0, the previewed curvature disturbance, and u_{prev}.
5: Solve the QP with OSQP subject to the steering-rate and steering boxes.
6: **if** solver fails **then** $u \leftarrow u_{\text{prev}}$ ▷ safe fallback
7: **else** $u \leftarrow$ first element of the optimal input sequence
8: **end if**
9: $u_{\text{prev}} \leftarrow u$
10: $\delta \leftarrow u$ (forward) **or** apply inner map (A.14) to obtain δ_f (reverse).
11: **return** steering rate from (A.2).

Only the first input of the optimal sequence is applied (receding horizon). The solved input is retained as the warm start for the next step and, like the PID integrator, is reset at episode boundaries. If the solver fails to converge, the controller falls back to the previous command.

A.6 Gain-tuning methodology

The controllers above are defined by their gains and weights: the proportional and feed-forward gains of pure pursuit, the PID gains, the LQR state/input weights, and the MPC horizon and cost weights. Rather than report a specific set of numbers, which are particular to one vehicle configuration, the point of interest is the procedure used to obtain them, which is the same for every controller.

Each controller is tuned by a bounded random search (a gain sweep) run directly against the simulator [93]. On each iteration a candidate gain set is sampled uniformly from hand-set ranges, the controller is rolled out over a batch of parallel environments, and the candidate is scored by the lexicographic key

$$\big(\text{completion rate}, \ -\text{mean cross-track error}\big), \tag{A.16}$$

so that reliably completing the path is maximized first and, among gains that complete, the one with the tightest tracking is preferred. The best candidate is retained. This search is run separately for each controller and each direction, because the forward and reverse plants differ so strongly.

The reverse variants in particular could not simply inherit the gains tuned on the small laboratory vehicle. On the shunt truck those gains destabilize the vehicle, and several gains (notably the hitch-stabilizing terms) change sign. This is a direct consequence of the open-loop-unstable reverse hitch dynamics and the large trailer-to-tractor length ratio $L_2/L_1 \approx 4.2$ discussed in appendix A.1.2. The reverse controllers were therefore re-tuned from scratch for the target geometry.

A.7 Evaluation harness

All four controllers are evaluated by a single harness that reuses the same batched environment as the reinforcement-learning policies, so that the classical baselines and the learned policy are measured on identical paths under identical dynamics. The harness runs both driving directions and, for each controller, records three per-episode metrics:

- **Completion**, the fraction of episodes that reach the goal (the mean success flag);
- **Trailer cross-track error**, the mean absolute lateral error of the trailer over the episode, a measure of tracking accuracy;
- **Peak hitch angle**, the largest articulation angle reached, a measure of how close the vehicle came to jackknifing (the jackknife threshold is 1.4 rad).

The harness handles the stateful controllers by passing a per-environment termination mask into the controller at each step, so the PID integrator (A.7) and the MPC warm start are reset exactly when an environment restarts. Evaluating every controller in the same loop, on the same feasible path pool, is what makes the reinforcement-learning comparison in the results chapter a fair, like-for-like one. The measured numbers are reported there rather than in this appendix.

Appendix B

Full Hyperparameter Tables

Table B.1: TD3 Hyperparameters

[Omitted for review]
Table content omitted for review.

Appendix C

Additional Plots

[Section content omitted for review.]

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